

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 95 on 18/12/2025. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMTgwODM2MTVfMzcyMw%3D%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

Table of Contents

1. [Introduction](#)
2. [Properties](#)
3. [Genome-wide overview](#)
4. [Most significant pathways](#)
5. [Pathways details](#)
6. [Identifiers found](#)
7. [Identifiers not found](#)

1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

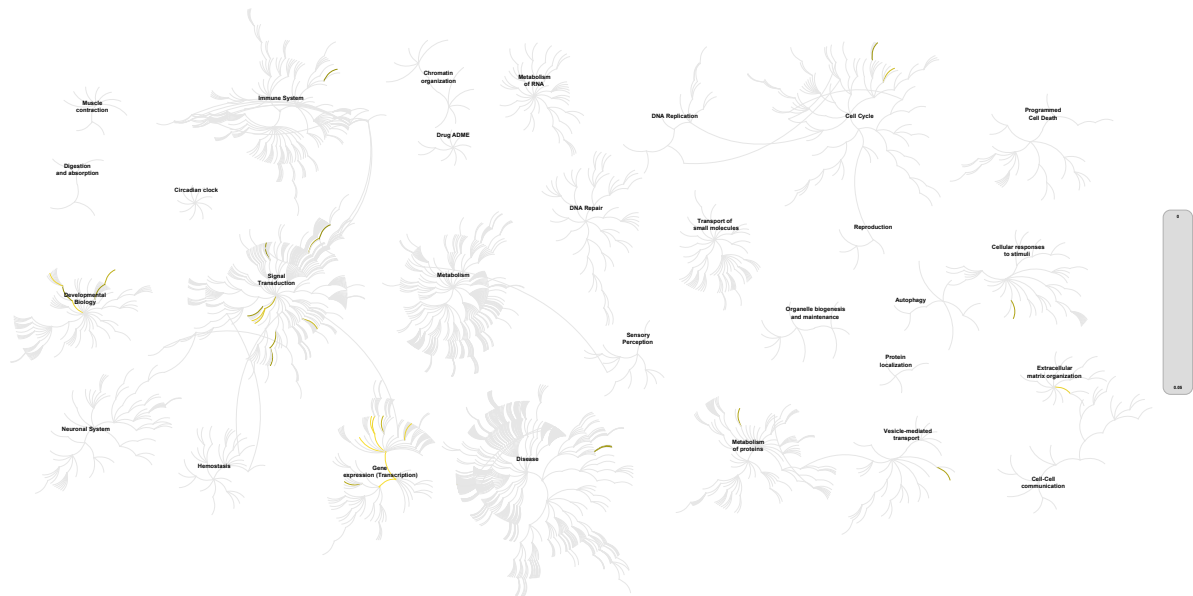
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 35 out of 59 identifiers in the sample were found in Reactome, where 321 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTEyMTgwODM2MTVfMzcyMw%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Nuclear Receptor transcription pathway	9 / 86	0.005	4.02e-10	1.32e-07	2 / 2	1.26e-04
Transcriptional Regulation by NPAS4	5 / 60	0.004	1.12e-05	0.002	6 / 46	0.003
Generic Transcription Pathway	20 / 1,574	0.097	3.59e-05	0.004	41 / 879	0.055
RNA Polymerase II Transcription	20 / 1,715	0.106	1.19e-04	0.01	41 / 940	0.059
Gene expression (Transcription)	21 / 1,958	0.121	2.50e-04	0.016	51 / 1,139	0.072
NPAS4 regulates expression of target genes	3 / 41	0.003	0.001	0.056	2 / 35	0.002
Regulation of NPAS4 gene expression	2 / 22	0.001	0.005	0.188	4 / 11	6.93e-04
MITF-M-regulated melanocyte development	5 / 233	0.014	0.005	0.188	8 / 150	0.009
Regulation of MITF-M-dependent genes involved in extracellular matrix, focal adhesion and epithelial-to-mesenchymal transition	2 / 23	0.001	0.006	0.188	2 / 22	0.001
RUNX1 regulates genes involved in megakaryocyte differentiation and platelet function	3 / 76	0.005	0.006	0.188	6 / 33	0.002
Non-integrin membrane-ECM interactions	3 / 83	0.005	0.007	0.217	6 / 33	0.002
mTORC1-mediated signalling	2 / 28	0.002	0.008	0.218	2 / 8	5.04e-04
Energy dependent regulation of mTOR by LKB1-AMPK	2 / 36	0.002	0.013	0.284	2 / 7	4.41e-04
EML4 and NUDC in mitotic spindle formation	3 / 121	0.007	0.02	0.284	3 / 5	3.15e-04
Deactivation of the beta-catenin transactivating complex	2 / 46	0.003	0.021	0.284	3 / 14	8.82e-04
TP53 Regulates Metabolic Genes	3 / 128	0.008	0.024	0.284	3 / 34	0.002
MTOR signalling	2 / 50	0.003	0.024	0.284	14 / 29	0.002
Somitogenesis	2 / 53	0.003	0.027	0.284	2 / 19	0.001
MPS VI - Maroteaux-Lamy syndrome	1 / 6	3.72e-04	0.028	0.284	2 / 2	1.26e-04
IRS activation	1 / 7	4.34e-04	0.033	0.284	3 / 3	1.89e-04
Post-transcriptional silencing by small RNAs	1 / 7	4.34e-04	0.033	0.284	2 / 3	1.89e-04
E3 ubiquitin ligases ubiquitinate target proteins	2 / 60	0.004	0.034	0.284	4 / 16	0.001
Golgi Associated Vesicle Biogenesis	2 / 61	0.004	0.035	0.284	5 / 7	4.41e-04

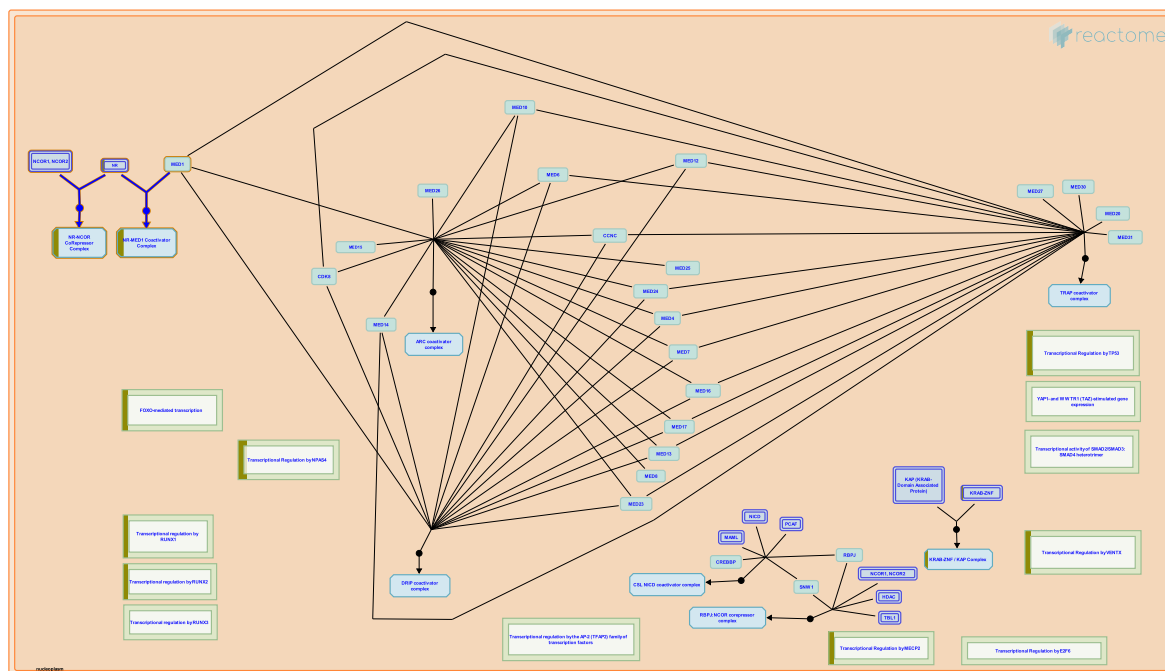
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Amino acids regulate mTORC1	2 / 61	0.004	0.035	0.284	8 / 12	7.56e-04
Toxicity of botulinum toxin type E (botE)	1 / 8	4.96e-04	0.037	0.284	3 / 5	3.15e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Nuclear Receptor transcription pathway (R-HSA-383280)



A classic example of bifunctional transcription factors is the family of Nuclear Receptor (NR) proteins. These are DNA-binding transcription factors that bind certain hormones, vitamins, and other small, diffusible signaling molecules. The non-liganded NRs recruit specific corepressor complexes of the NCOR/SMRT type, to mediate transcriptional repression of the target genes to which they are bound. During signaling, ligand binding to a specific domain the NR proteins induces a conformational change that results in the exchange of the associated CoR complex, and its replacement by a specific coactivator complex of the TRAP / DRIP / Mediator type. These coactivator complexes typically nucleate around a MED1 coactivator protein that is directly bound to the NR transcription factor.

A general feature of the 49 human NR proteins is that in the unliganded state, they each bind directly to an NCOR corepressor protein, either NCOR1 or NCOR2 (NCOR2 was previously named "SMRT"). This NCOR protein nucleates the assembly of additional, specific corepressor proteins, depending on the cell and DNA context. The NR-NCOR interaction is mediated by a specific protein interaction domain (PID) present in the NRs that binds to specific cognate PID(s) present in the NCOR proteins. Thus, the human NRs each take part in an NR-NCOR binding reaction in the absence of binding by their ligand.

A second general feature of the NR proteins is that they each contain an additional, but different PID that mediates specific binding interactions with MED1 proteins. In the ligand-bound state, NRs each take part in an NR-MED1 binding reaction to form an NR-MED1 complex. The bound MED1 then functions to nucleate the assembly of additional specific coactivator proteins, depending on the cell and DNA context, such as what specific target gene promoter they are bound to, and in what cell type.

The formation of specific MED1-containing coactivator complexes on specific NR proteins has been well-characterized for a number of the human NR proteins (see Table 1 in (Bourbon, 2004)). For example, binding of thyroid hormone (TH) to the human TH Receptor (THRA or THRB) was found to result in the recruitment of a specific complex of Thyroid Receptor Associated Proteins - the TRAP coactivator complex - of which the TRAP220 subunit was later identified to be the Mediator 1 (MED1) homologue.

Similarly, binding of Vitamin D to the human Vitamin D3 Receptor was found to result in the recruitment of a specific complex of D Receptor Interacting Proteins - the DRIP coactivator complex, of which the DRIP205 subunit was later identified to be human MED1.

References

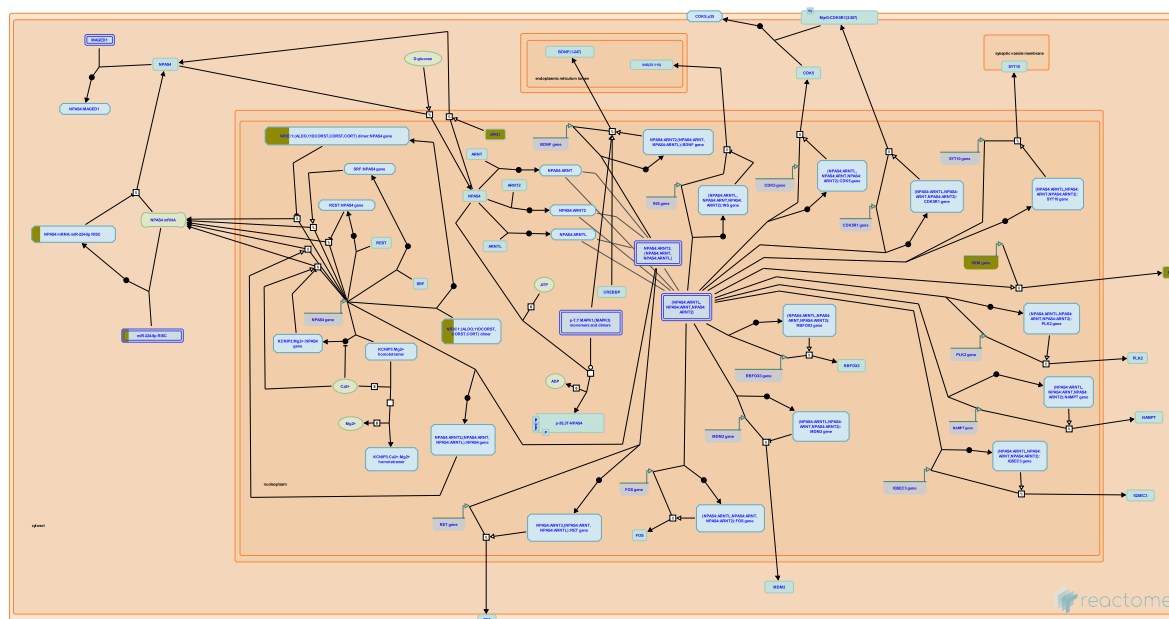
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Date	Action	Author
2008-11-20	Authored	Caudy M
2008-12-03	Created	Caudy M
2009-05-27	Edited	Caudy M
2009-08-29	Reviewed	Freedman LP

1 submitted entities found in this pathway, mapping to 9 Reactome entities

Input	UniProt Id
NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9

2. Transcriptional Regulation by NPAS4 (R-HSA-9634815)



NPAS4 (Neuronal PAS domain containing protein 4) is a calcium dependent transcription factor predominantly expressed in neurons that regulates activation of genes involved in neuronal circuit formation, function, and plasticity (Ooe et al. 2004; Lin et al. 2008; Ramamoorthi et al. 2011; Maya-Vetencourt 2013; Sun and Lin 2016; Weng et al. 2018). NPAS4 possesses a conserved basic helix loop helix (bHLH) motif and a PAS domain (Fahim et al. 2018). NPAS4 is among the most rapidly induced immediate early genes (IEGs), which are activated after sensory and behavioral experience and thought to be crucial for formation of long term memory (Ramamoorthi et al. 2011; Sun et al. 2016; Heslin and Coutellier 2018; Weng et al. 2018). NPAS4 is activated within minutes of neuronal stimulation to regulate the formation of inhibitory synapses (Lin et al. 2008). Transcriptional targets of NPAS4 include transcription factors and proteins involved in signal transduction and protein trafficking (Lin et al. 2008). NPAS4 regulates development of glutamatergic and GABAergic synapses essential for information processing and memory formation (Lin et al. 2008, Weng et al. 2018). NPAS4 induced gene expression programs differ between excitatory and inhibitory neurons (Spiegel et al. 2014), leading to a circuit wide homeostatic response. Besides directly regulating function of neurons, NPAS4 may be involved in the regulation of neuroinflammation and neuronal apoptosis (Zhang et al. 2009; Choy et al. 2015; Fan et al. 2016; Zhang et al. 2021). NPAS4 is expressed in the pancreatic beta cells and regulates their function under stress conditions (Sabatini et al. 2018). For review, please refer to Sun and Lin 2016, and Fu et al. 2020.

References

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- Fahim A, Rehman Z, Bhatti MF, Ali A, Virk N, Rashid A & Paracha RZ (2018). Structural insights and characterization of human Npas4 protein. *PeerJ*, 6, e4978. [🔗](#)

Heslin K & Coutellier L (2018). Npas4 deficiency and prenatal stress interact to affect social recognition in mice. *Genes Brain Behav*, 17, e12448. [🔗](#)

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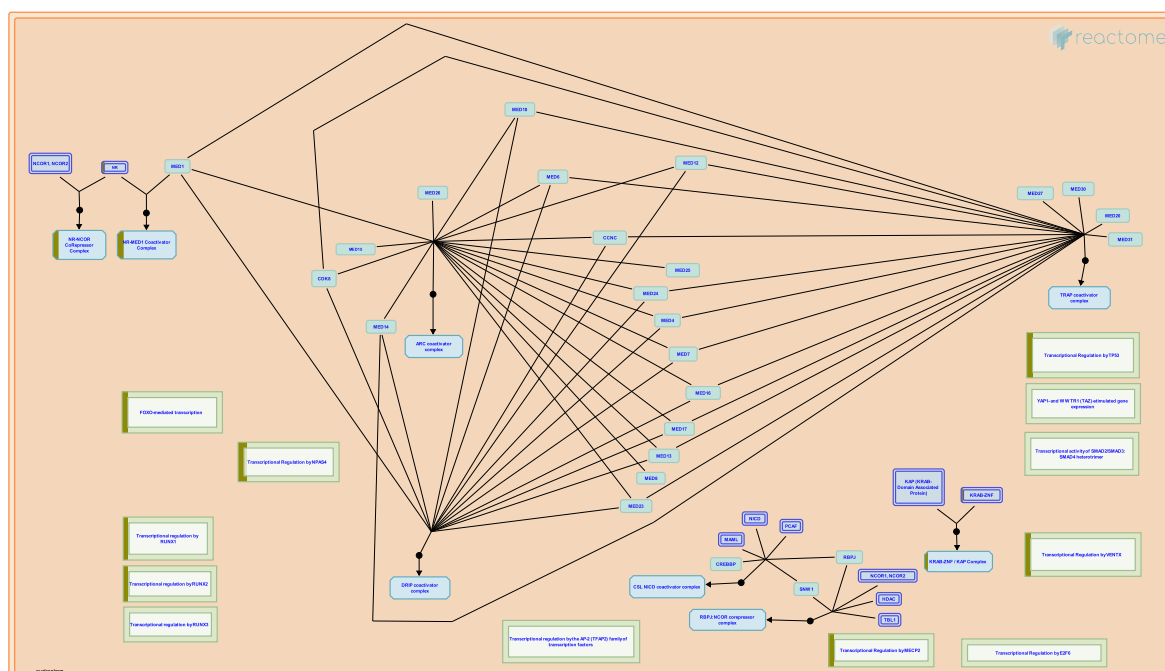
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2019-01-15	Created	Orlic-Milacic M
2022-04-08	Authored	Orlic-Milacic M
2022-07-29	Reviewed	Lin Y
2022-08-02	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

4 submitted entities found in this pathway, mapping to 5 Reactome entities

Input	UniProt Id	Input	UniProt Id
GEM	P55040	NR3C1	P04150
TNRC6B	Q9UPQ9	XPO1	O14980

Input	Ensembl Id
GEM	ENSG00000164949

3. Generic Transcription Pathway (R-HSA-212436)



OVERVIEW OF TRANSCRIPTION REGULATION:

Detailed studies of gene transcription regulation in a wide variety of eukaryotic systems has revealed the general principles and mechanisms by which cell- or tissue-specific regulation of differential gene transcription is mediated (reviewed in Naar, 2001. Kadonaga, 2004, Maston, 2006, Barolo, 2002; Roeder, 2005, Rosenfeld, 2006). Of the three major classes of DNA polymerase involved in eukaryotic gene transcription, Polymerase II generally regulates protein-encoding genes. Figure 1 shows a diagram of the various components involved in cell-specific regulation of Pol-II gene transcription.

Core Promoter: Pol II-regulated genes typically have a Core Promoter where Pol II and a variety of general factors bind to specific DNA motifs:

- i: the TATA box (TATA DNA sequence), which is bound by the "TATA-binding protein" (TBP).
- ii: the Initiator motif (INR), where Pol II and certain other core factors bind, is present in many Pol II-regulated genes.
- iii: the Downstream Promoter Element (DPE), which is present in a subset of Pol II genes, and where additional core factors bind.

The core promoter binding factors are generally ubiquitously expressed, although there are exceptions to this.

Proximal Promoter: immediately upstream (5') of the core promoter, Pol II target genes often have a Proximal Promoter region that spans up to 500 base pairs (b.p.), or even to 1000 b.p.. This region contains a number of functional DNA binding sites for a specific set of transcription activator (TA) and transcription repressor (TR) proteins. These TA and TR factors are generally cell- or tissue-specific in expression, rather than ubiquitous, so that the presence of their cognate binding sites in the proximal promoter region programs cell- or tissue-specific expression of the target gene, perhaps in conjunction with TA and TR complexes bound in distal enhancer regions.

Distal Enhancer(s): many or most Pol II regulated genes in higher eukaryotes have one or more distal Enhancer regions which are essential for proper regulation of the gene, often in a cell or tissue-specific pattern. Like the proximal promoter region, each of the distal enhancer regions typically contain a cluster of binding sites for specific TA and/or TR DNA-binding factors, rather than just a single site.

Enhancers generally have three defining characteristics:

- i: They can be located very long distances from the promoter of the target gene they regulate, sometimes as far as 100 Kb, or more.
- ii: They can be either upstream (5') or downstream (3') of the target gene, including within introns of that gene.
- iii: They can function in either orientation in the DNA.

Combinatorial mechanisms of transcription regulation: The specific combination of TA and TR binding sites within the proximal promoter and/or distal enhancer(s) provides a "combinatorial transcription code" that mediates cell- or tissue-specific expression of the associated target gene. Each promoter or enhancer region mediates expression in a specific subset of the overall expression pattern. In at least some cases, each enhancer region functions completely independently of the others, so that the overall expression pattern is a linear combination of the expression patterns of each of the enhancer modules.

Co-Activator and Co-Repressor Complexes: DNA-bound TA and TR proteins typically recruit the assembly of specific Co-Activator (Co-A) and Co-Repressor (Co-R) Complexes, respectively, which are essential for regulating target gene transcription. Both Co-A's and Co-R's are multi-protein complexes that contain several specific protein components.

Co-Activator complexes generally contain at least one component protein that has Histone Acetyltransferase (HAT) enzymatic activity. This functions to acetylate Histones and/or other chromatin-associated factors, which typically increases that transcription activation of the target gene. By contrast, **Co-Repressor complexes** generally contain at least one component protein that has Histone De-Acetylase (HDAC) enzymatic activity. This functions to de-acetylate Histones and/or other chromatin-associated factors. This typically increases the transcription repression of the target gene.

Adaptor (Mediator) complexes: In addition to the co-activator complexes that assemble on particular cell-specific TA factors, - there are at least two additional transcriptional co-activator complexes common to most cells. One of these is the Mediator complex, which functions as an "adaptor" complex that bridges between the tissue-specific co-activator complexes assembled in the proximal promoter (or distal enhancers). The human Mediator complex has been shown to contain at least 19 protein distinct components. Different combinations of these co-activator proteins are also found to be components of specific transcription Co-Activator complexes, such as the DRIP, TRAP and ARC complexes described below.

TBP/TAF complex: Another large Co-A complex is the "TBP-associated factors" (TAFs) that assemble on TBP (TATA-Binding Protein), which is bound to the TATA box present in many promoters. There are at least 23 human TAF proteins that have been identified. Many of these are ubiquitously expressed, but TAFs can also be expressed in a cell or tissue-specific pattern.

Specific Coactivator Complexes for DNA-binding Transcription Factors.

A number of specific co-activator complexes for DNA-binding transcription factors have been identified, including DRIP, TRAP, and ARC (reviewed in Bourbon, 2004, Blazek, 2005, Conaway, 2005, and Malik, 2005). The DRIP co-activator complex was originally identified and named as a specific complex associated with the Vitamin D Receptor member of the nuclear receptor family of transcription factors (Rachez, 1998). Similarly, the TRAP co-activator complex was originally identified as a complex that associates with the thyroid receptor (Yuan, 1998). It was later determined that all of the components of the DRIP complex are also present in the TRAP complex, and the ARC complex (discussed further below). For example, the DRIP205 and TRAP220 proteins were shown to be identical, as were specific pairs of the other components of these complexes (Rachez, 1999).

In addition, these various transcription co-activator proteins identified in mammalian cells were found to be the orthologues or homologues of the Mediator ("adaptor") complex proteins (reviewed in Bourbon, 2004). The Mediator proteins were originally identified in yeast by Kornberg and colleagues, as complexes associated with DNA polymerase (Kelleher, 1990). In higher organisms, Adapter complexes bridge between the basal transcription factors (including Pol II) and tissue-specific transcription factors (TFs) bound to sites within upstream Proximal Promoter regions or distal Enhancer regions (Figure 1). However, many of the Mediator homologues can also be found in complexes associated with specific transcription factors in higher organisms. A unified nomenclature system for these adapter / co-activator proteins now labels them Mediator 1 through Mediator 31 (Bourbon, 2004). For example, the DRIP205 / TRAP220 proteins are now identified as Mediator 1 (Rachez, 1999), based on homology with yeast Mediator 1.

Example Pathway: Specific Regulation of Target Genes During Notch Signaling:

One well-studied example of cell-specific regulation of gene transcription is selective regulation of target genes during Notch signaling. Notch signaling was first identified in *Drosophila*, where it has been studied in detail at the genetic, molecular, biochemical and cellular levels (reviewed in Justice, 2002; Bray, 2006; Schweisguth, 2004; Louvri, 2006). In *Drosophila*, Notch signaling to the nucleus is thought always to be mediated by one specific DNA binding transcription factor, Suppressor of Hairless. In mammals, the homologous genes are called CBF1 (or RBPJ κ), while in worms they are called Lag-1, so that the acronym "CSL" has been given to this conserved transcription factor family. There are at least two human CSL homologues, which are now named RBPJ and RBPJL.

In *Drosophila*, Su(H) is known to be bifunctional, in that it represses target gene transcription in the absence of Notch signaling, but activates target genes during Notch signaling. At least some of the mammalian CSL homologues are believed also to be bifunctional, and to mediate target gene repression in the absence of Notch signaling, and activation in the presence of Notch signaling.

Notch Co-Activator and Co-Repressor complexes: This repression is mediated by at least one specific co-repressor complexes (Co-R) bound to CSL in the absence of Notch signaling. In *Drosophila*, this co-repressor complex consists of at least three distinct co-repressor proteins: Hairless, Groucho, and dCtBP (*Drosophila* C-terminal Binding Protein). Hairless has been shown to bind directly to Su(H), and Groucho and dCtBP have been shown to bind directly to Hairless (Barolo, 2002). All three of the co-repressor proteins have been shown to be necessary for proper gene regulation during Notch signaling in vivo (Nagel, 2005).

In mammals, the same general pathway and mechanisms are observed, where CSL proteins are bi-functional DNA binding transcription factors (TFs), that bind to Co-Repressor complexes to mediate repression in the absence of Notch signaling, and bind to Co-Activator complexes to mediate activation in the presence of Notch signaling. However, in mammals, there may be multiple co-repressor complexes, rather than the single Hairless co-repressor complex that has been observed in *Drosophila*.

During Notch signaling in all systems, the Notch transmembrane receptor is cleaved and the Notch intracellular domain (NICD) translocates to the nucleus, where it there functions as a specific transcription co-activator for CSL proteins. In the nucleus, NICD replaces the Co-R complex bound to CSL, thus resulting in de-repression of Notch target genes in the nucleus (Figure 2). Once bound to CSL, NICD and CSL proteins recruit an additional co-activator protein, Mastermind, to form a CSL-NICD-Mam ternary co-activator (Co-A) complex. This Co-R complex was initially thought to be sufficient to mediate activation of at least some Notch target genes. However, there now is evidence that still other co-activators and additional DNA-binding transcription factors are required in at least some contexts (reviewed in Barolo, 2002).

Thus, CSL is a good example of a bifunctional DNA-binding transcription factor that mediates repression of specific targets genes in one context, but activation of the same targets in another context. This bifunctionality is mediated by the association of specific Co-Repressor complexes vs. specific Co-Activator complexes in different contexts, namely in the absence or presence of Notch signaling.

References

- Näär AM, Lemon BD & Tjian R (2001). Transcriptional coactivator complexes. *Annu Rev Biochem*, 70, 475-501. [↗](#)
- Kadonaga JT (2004). Regulation of RNA polymerase II transcription by sequence-specific DNA binding factors. *Cell*, 116, 247-57. [↗](#)
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- Barolo S & Posakony JW (2002). Three habits of highly effective signaling pathways: principles of transcriptional control by developmental cell signaling. *Genes Dev*, 16, 1167-81. [↗](#)
- Roeder RG (2005). Transcriptional regulation and the role of diverse coactivators in animal cells. *FEBS Lett*, 579, 909-15. [↗](#)

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Date	Action	Author
2008-02-09	Created	Caudy M
2008-02-26	Reviewed	Freedman LP
2025-11-15	Modified	Weiser JD

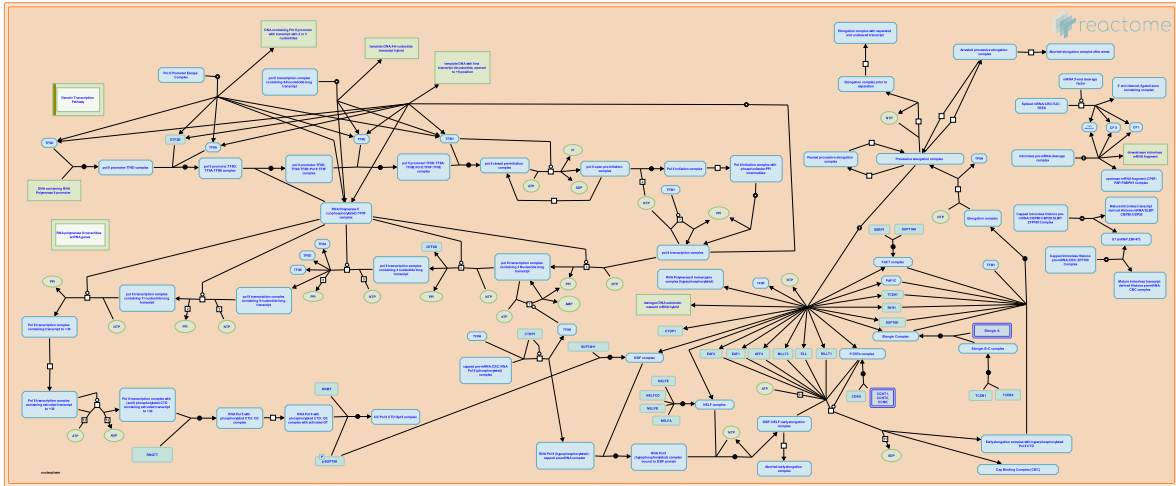
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THBS1	P07996	TNRC6B	Q9UPQ9	XPO1	O14980
ZNF268	Q14587, Q96BR6				

Input	Ensembl Id	Input	Ensembl Id
GEM	ENSG00000164949	THBS1	ENSG00000137801

4. RNA Polymerase II Transcription (R-HSA-73857)



Cellular compartments: nucleoplasm.

RNA polymerase II (Pol II) is the central enzyme that catalyses DNA- directed mRNA synthesis during the transcription of protein-coding genes. Pol II consists of a 10-subunit catalytic core, which alone is capable of elongating the RNA transcript, and a complex of two subunits, Rpb4/7, that is required for transcription initiation.

The transcription cycle is divided in three major phases: initiation, elongation, and termination. Transcription initiation include promoter DNA binding, DNA melting, and initial synthesis of short RNA transcripts. The transition from initiation to elongation, is referred to as promoter escape and leads to a stable elongation complex that is characterized by an open DNA region or transcription bubble. The bubble contains the DNA-RNA hybrid, a heteroduplex of eight to nine base pairs. The growing 3-end of the RNA is engaged with the polymerase complex active site. Ultimately transcription terminates and Pol II dissociates from the template.

References

Cramer P (2004). Structure and function of RNA polymerase II. Adv. Protein Chem., 67, 1-42. [🔗](#)

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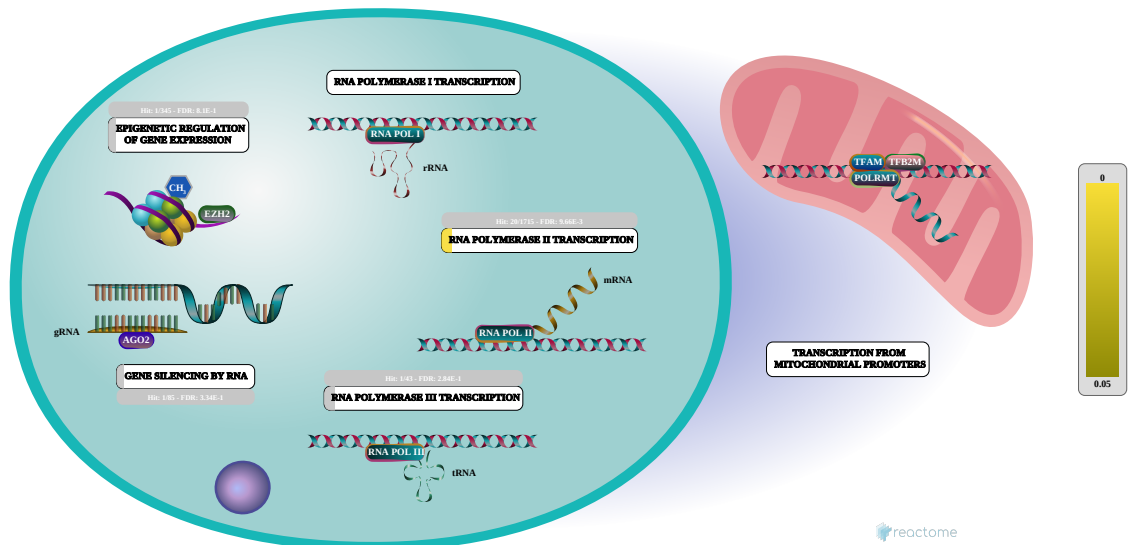
Date	Action	Author
2005-01-12	Created	Reinberg D, Timmers HTM, Conaway JW, Proudfoot NJ, Conaway RC
2025-11-15	Modified	Weiser JD

7 submitted entities found in this pathway, mapping to 20 Reactome entities

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THBS1	P07996	TNRC6B	Q9UPQ9	XPO1	O14980
ZNF268	Q14587, Q96BR6				

Input	Ensembl Id	Input	Ensembl Id
GEM	ENSG00000164949	THBS1	ENSG00000137801

5. Gene expression (Transcription) (R-HSA-74160)



Gene expression encompasses transcription and translation and the regulation of these processes. RNA Polymerase I Transcription produces the large preribosomal RNA transcript (45S pre-rRNA) that is processed to yield 18S rRNA, 28S rRNA, and 5.8S rRNA, accounting for about half the RNA in a cell. RNA Polymerase II transcription produces messenger RNAs (mRNA) as well as a subset of non-coding RNAs including many small nucleolar RNAs (snRNA) and microRNAs (miRNA). RNA Polymerase III Transcription produces transfer RNAs (tRNA), 5S RNA, 7SL RNA, and U6 snRNA. Transcription from mitochondrial promoters is performed by the mitochondrial RNA polymerase, POLRMT, to yield long transcripts from each DNA strand that are processed to yield 12S rRNA, 16S rRNA, tRNAs, and a few RNAs encoding components of the electron transport chain. Regulation of gene expression can be divided into epigenetic regulation, transcriptional regulation, and post-transcription regulation (comprising translational efficiency and RNA stability). Epigenetic regulation of gene expression is the result of heritable chemical modifications to DNA and DNA-binding proteins such as histones. Epigenetic changes result in altered chromatin complexes that influence transcription. Gene Silencing by RNA mostly occurs post-transcriptionally but can also affect transcription. Small RNAs originating from the genome (miRNAs) or from exogenous RNA (siRNAs) are processed and transferred to the RNA-induced silencing complex (RISC), which interacts with complementary RNA to cause cleavage, translational inhibition, or transcriptional inhibition.

References

Edit history

Date	Action	Author
2003-09-11	Authored	Larsson NG, Comai L, Gustafsson CM, Reinberg D, Timmers HTM et al.
2003-09-11	Created	Proudfoot NJ, Kornblihtt AR
2008-12-03	Authored	Caudy M, Proudfoot NJ, Kornblihtt AR, D'Eustachio P
2016-12-29	Revised	D'Eustachio P
2025-11-12	Edited	Joshi-Tope G

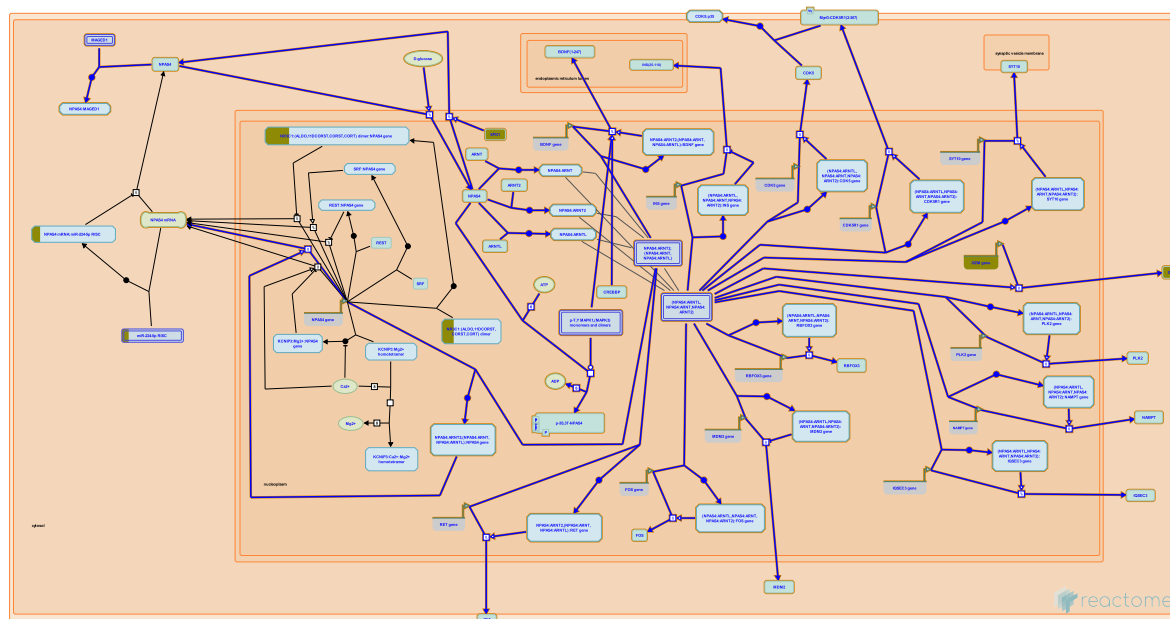
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2025-11-12	Reviewed	Paule M, Willis I, Zhao X
2025-11-15	Modified	Weiser JD

8 submitted entities found in this pathway, mapping to 21 Reactome entities

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RRAGA	Q5VZM2, Q7L523	THBS1	P07996	TNRC6B	Q9UPQ9
XPO1	O14980	ZNF268	Q14587, Q96BR6		

Input	Ensembl Id	Input	Ensembl Id
GEM	ENSG00000164949	THBS1	ENSG00000137801

6. NPAS4 regulates expression of target genes (R-HSA-9768919)



Cellular compartments: nucleoplasm.

NPAS4 is a basic helix loop helix (bHLH) transcription factor that needs to dimerize with another bHLH protein, either ARNT, ARNT2 or ARNTL, in order to be able to bind to target DNA (Ooe et al. 2004; Ooe et al. 2009).

NPAS4 is implicated as a transcriptional regulator of genes involved in neuronal development such as CDK5 (Yun et al. 2013), CDK5R1 (Yun et al. 2013), RBFOX3 (NeuN) (Yun et al. 2013), BDNF (Pruunsild et al. 2011) and RET (Sribudiani et al. 2011), genes involved in synaptogenesis and synaptic transmission such as NPTX2 (Lin et al. 2008), MDM2 (Yoshihara et al. 2014; Lv et al. 2021), FOS (Ramamoorthi et al. 2011), IQSEC3 (Kim et al. 2021), PLK2 (Weng et al. 2018) and possibly other genes (Lin et al. 2008; Shan et al. 2018), circadian rhythm-related genes such as NAMPT (West et al. 2013), and genes involved in neuroprotection upon injury such as GEM (Takahashi et al. 2021), SYT10 (Woitecki et al. 2016) and possibly other genes (Qiu et al. 2013). In pancreatic beta-cell, NPAS4 is implicated as a regulator of insulin synthesis under stress conditions (Sabatini et al. 2013).

The circadian clock regulated gene CRY1 was identified as NPAS4 target gene in sheep brain (West et al. 2013). The DBNL gene, encoding Drebrin, a dendritic cytoskeleton modulator, was identified as a gene directly upregulated by Npas4 (Ooe et al. 2004)

NPAS4 is expressed in endothelial cells and may play a role in angiogenesis (Esser et al. 2017).

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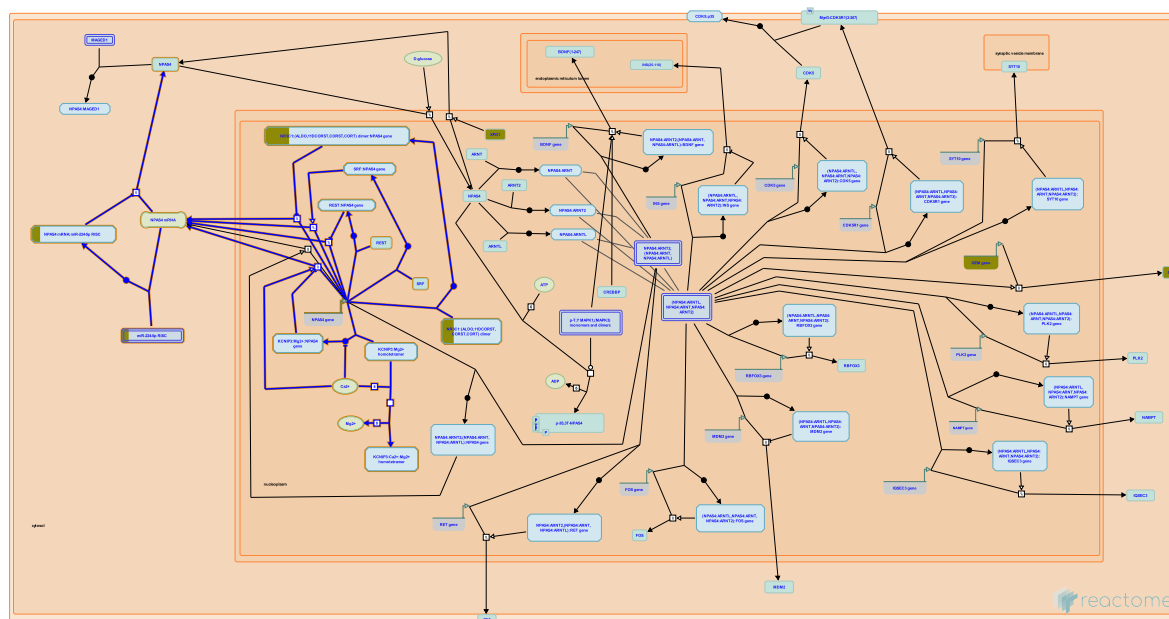
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2022-07-29	Reviewed	Lin Y
2022-08-02	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 3 Reactome entities

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GEM	P55040	XPO1	O14980

Input	Ensembl Id
GEM	ENSG00000164949

7. Regulation of NPAS4 gene expression ([R-HSA-9768759](#))



NPAS4 is predominantly expressed in neuronal cells. In neurons, expression of NPAS4 is regulated by Ca^{2+} , which ties NPAS4 level to neuronal activity (Lin et al. 2008; Zhang et al. 2009; Kim et al. 2010; Mellström et al. 2014; Lobos et al. 2021).

NPAS4 expression is regulated by signaling pathways activated downstream of the N-methyl-D-aspartate (NMDA) receptor and L-type Ca^{2+} channel (Coba et al. 2008; Lin et al. 2008; Horvath et al. 2021).

In vivo, NPAS4 expression is activated by visual stimulation (Lin et al. 2008), contextual learning (Ramamoorthi et al. 2011), as well as pharmacologically induced generalized seizure (Flood et al. 2004). Chronic stress induces downregulation of NPAS4 gene expression in the hippocampus (Furukawa-Hibi et al. 2015), so does social isolation (Ibi et al. 2008). Adult female mice who were exposed to early maternal separation (early life stress) show upregulation of NPAS4 in the prefrontal cortex later in life (Ryabushkina et al. 2020). NPAS4 is thought to have a neuroprotective effect and is downregulated during neurodegeneration (Ooe et al. 2009; Zhang et al. 2009; Louis Sam Titus et al. 2019). NPAS4 level is reduced in the hippocampus of aged, memory-impaired mice (Qiu et al. 2016). Studies in mice indicate that NPAS4 affects synaptic connections in excitatory and inhibitory neurons, neural circuit plasticity, and memory formation (reviewed by Sun and Lin 2016). NPAS4 may be involved in the functioning of the circadian system (Unfried et al. 2010; West et al. 2013; Xu et al. 2021).

Npas4 mRNA levels are downregulated upon infection with Zika virus through an unknown mechanism (Alpuche-Lazcano et al. 2021).

Besides neuronal cells, NPAS4 is also expressed in pancreatic beta cells, where its levels are regulated by intracellular calcium, as in the nervous system (Speckmann et al. 2016).

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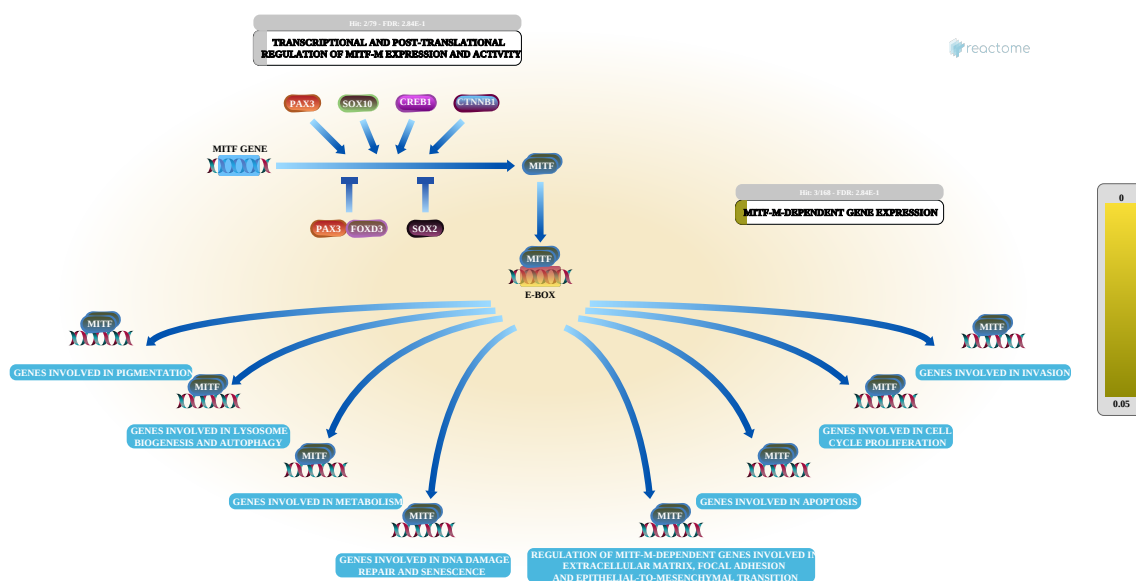
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2022-07-29	Reviewed	Lin Y
2022-08-02	Edited	Orlic-Milacic M
2023-03-08	Modified	Matthews L

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
NR3C1	P04150	TNRC6B	Q9UPQ9

8. MITF-M-regulated melanocyte development (R-HSA-9730414)



Melanocyte Inducing Factor (MITF, also known as Microphthalmia-associated transcription factor) is a key regulator of melanocyte differentiation and development during embryogenesis, of differentiation of melanocyte stem cells post-natally, and of melanoma cells.

Melanocytes are cells that possess specialized organelles called melanosomes that synthesize eumelanin and pheomelanin from tyrosine in a series of reactions. Melanosomes are transferred from cutaneous melanocytes to adjacent keratinocytes to provide protection against UV as well as coloration of skin, eye, hair, feathers and scales. Besides being found in the basal layer of the skin, melanocytes are also present in hair follicles, the inner ear and in the iris eye, among other places. The eye also contains a layer of melanosome-containing cells behind the retina, called the retinal pigment epithelium (RPE) (reviewed in Mort et al, 2015; D'Mello et al, 2016; Goding and Arnheiter, 2019; Le et al, 2021; Cui and Man, 2023).

Cutaneous melanocytes and their precursors, melanoblasts, arise during embryogenesis from neural crest cells that migrate dorsolaterally through the developing embryo (reviewed in Mort et al, 2015). They also arise from glial/melanoblast precursors migrating on a ventromedial pathway and along nerves (Adameyko et al, 2009). Expression of MITF is a key determinant of melanocyte fate, and mutations in MITF are associated with a variety of defects in pigmentation as well as with deafness (due to absence of melanocytes in the inner ear) and microphthalmia (due to aberrant development of retina and RPE), among other conditions (reviewed in White and Zon, 2008; Mort et al, 2015; Goding and Arnheiter, 2019; Le et al, 2021).

The gene for MITF encodes several distinct isoforms based on alternative splicing. The gene has a 3' portion consisting of exons 2-9 that are generally shared by all transcripts. In mice and humans, the upstream region of the gene contains 9 exons, some of them coding, some not, and each regulated by its own promoter. Most of them are spliced to exon 2 via a common exon 1B. An exception is exon 1M which is directly spliced to exon 2, giving rise to the so-called M-isoform of MITF. This arrangement gives rise to a number of different mRNA and protein isoforms with preferential expression patterns. Exon 1A-containing transcripts, for instance, are ubiquitously expressed, exon 1H-containing transcripts are highly expressed in the heart, exon 1D-containing transcripts are expressed in the RPE, and exon 1M-containing transcripts are expressed in neural crest-derived melanocytes. Nevertheless, there is little information on whether the different isoforms have different functions except that exon 1B-containing transcripts (but not MITF-M) harbor a sequence subject to mTORC1 regulation (reviewed in Goding and Arnheiter, 2019; Vu et al, 2020). Most if not all transcripts come in two additional splice versions, one including and one excluding 18 bp of part of exon 6, called exon 6A, which encodes 6 amino acids lying upstream of the DNA-binding domain and which is regulated by MAPK signaling (Primot et al, 2010). They are usually referred to as the (+) and (-) versions of MITF. While the (-) version of a fragment of MITF-M has slightly reduced DNA-binding affinity compared to the (+) version, no specific role has so far been found for exon 6A (Pogenburg et al, 2012). Cell-based assays suggest that MITF (+) has a strong inhibitory effect on cellular proliferation relative to the (-) version, and MITF (-) is expressed at high levels in melanoma cells (Bismuth et al, 2005; Primot et al, 2010).

This pathway focuses on the activity of the melanocyte lineage-specific transcription factor MITF-M, although some of the biology described may also be relevant for other MITF isoforms.

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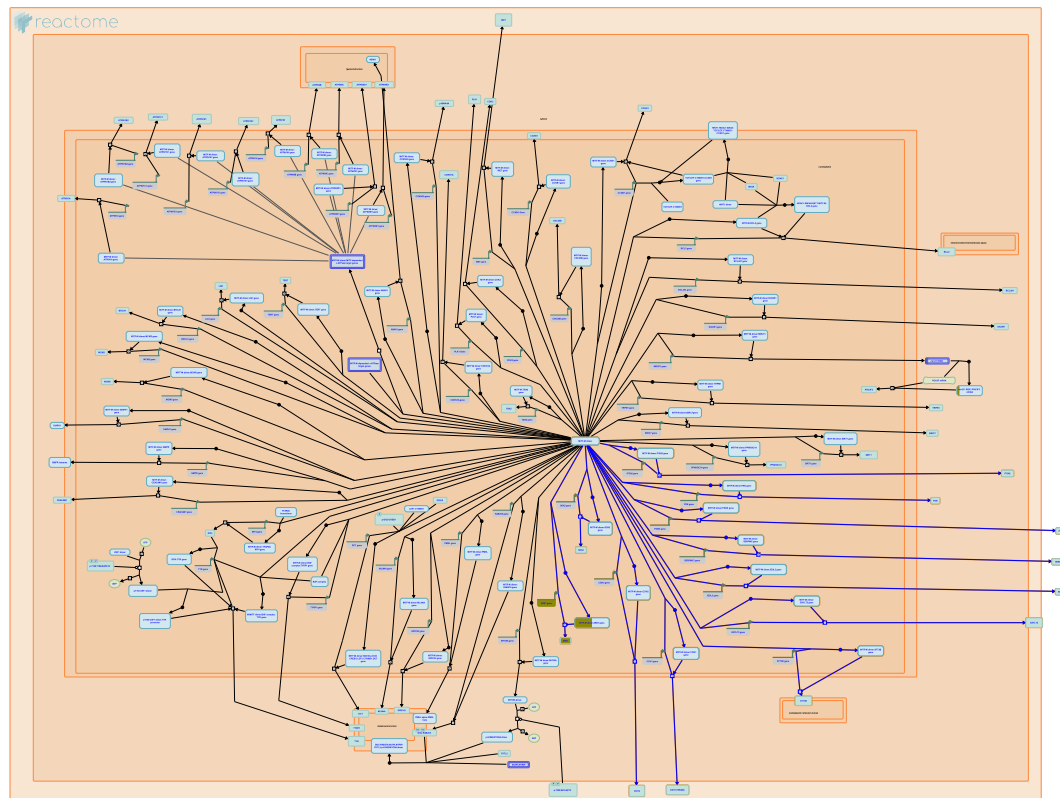
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2024-02-26	Reviewed	Arnheiter H
2024-05-21	Reviewed	de la Serna IL
2024-11-05	Reviewed	Vu HN, Steingrímsson E
2025-11-15	Modified	Weiser JD

4 submitted entities found in this pathway, mapping to 5 Reactome entities

Input	UniProt Id	Input	UniProt Id
CSF1	P09603	TNRC6B	Q9UPQ9
XPO1	O14980	ZEB1	P37275

Input	Ensembl Id
ZEB1	ENSG00000148516

9. Regulation of MITF-M-dependent genes involved in extracellular matrix, focal adhesion and epithelial-to-mesenchymal transition ([R-HSA-9926550](#))



Studies in melanoma cells have identified MITF-M as a regulator of targets involved in extracellular matrix organization (ECM), focal adhesion (FA) and epithelial-to-mesenchymal transition (EMT) (Dilshat et al, 2021). siRNA-mediated knockdown of MITF-M in the SKMel28 melanoma cell line causes morphological and cytoskeletal changes and promotes formation of aggregates on matrigel, consistent with a role for MITF-M in ECM organization and EMT. CUT-and-RUN analysis identified 101 genes associated with ECM and FA organization that are directly bound by MITF-M and that show increased expression upon siRNA MITF-knockdown (Dilshat et al, 2021), implicating MITF-M as a repressor of these targets in melanoma cells. Of these, 42 were additionally identified as direct MITF-M targets by ChIP-seq in two independent melanoma cell line studies (Laurette et al, 2015; Webster et al, 2014). MITF-M expression levels are correlated with different cellular phenotypes, with high levels corresponding to a more proliferative phenotype and low levels to a quiescent/invasive phenotype, EMT and drug resistance (reviewed in Rambow et al, 2019). Consistent with a role for MITF-M in phenotypic change, EMT-related genes such as CDH1 (E-cadherin), CDH2 (N-cadherin), SOX2 and ZEB1 were also identified as direct MITF-M targets (Dilshat et al, 2021).

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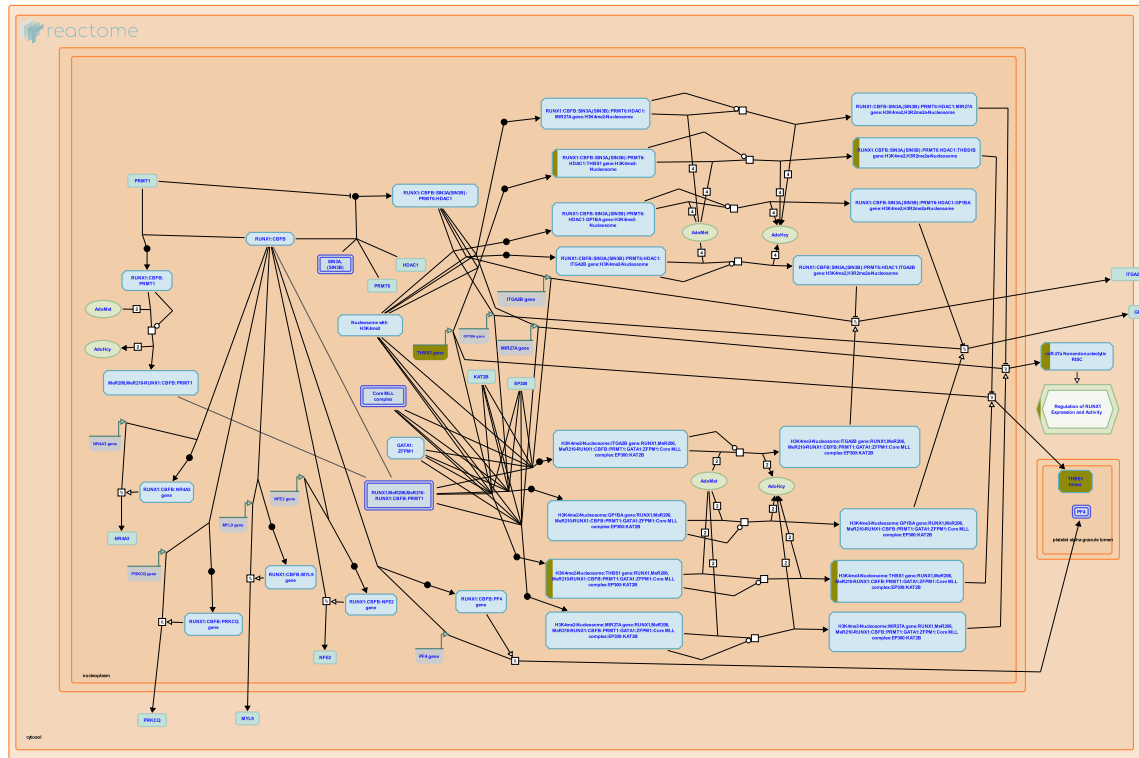
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2024-11-05	Authored	Vu HN, Steingrímsson E
2024-11-08	Modified	Rothfels K

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
ZEB1	P37275

Input	Ensembl Id
ZEB1	ENSG00000148516

10. RUNX1 regulates genes involved in megakaryocyte differentiation and platelet function (R-HSA-8936459)



In human hematopoietic progenitors, RUNX1 and its partner CBFB are up-regulated at the onset of megakaryocytic differentiation and down-regulated at the onset of erythroid differentiation. The complex of RUNX1 and CBFB cooperates with the transcription factor GATA1 in the transactivation of megakaryocyte-specific genes. In addition, RUNX1 and GATA1 physically interact (Elagib et al. 2003), and this interaction involves the zinc finger domain of GATA1 (Xu et al. 2006). Other components of the RUNX1:CBFB activating complex at megakaryocytic promoters are GATA1 heterodimerization partner, ZFP101 (FOG1), histone acetyltransferases EP300 (p300) and KAT2B (PCAF), the WDR5-containing histone methyltransferase MLL complex and the arginine methyltransferase PRMT1 (Herglotz et al. 2013). In the absence of PRMT1, the transcriptional repressor complex can form at megakaryocytic promoters, as RUNX1 that is not arginine methylated can bind to SIN3A/SIN3B co-repressors (Zhao et al. 2008). Besides SIN3A/SIN3B, the RUNX1:CBFB repressor complex at megakaryocytic promoters also includes histone deacetylase HDAC1 and histone arginine methyltransferase PRMT6 (Herglotz et al. 2013).

Megakaryocytic promoters regulated by the described RUNX1:CBFB activating and repressing complexes include ITGA2B, GP1BA, THBS1 and MIR27A (Herglotz et al. 2013). ITGA2B is only expressed in maturing megakaryocytes and platelets and is involved in platelet aggregation (Block and Poncz 1995). GP1BA is expressed at the cell surface membrane of maturing megakaryocytes and platelets and participates in formation of platelet plugs (Cauwenberghs et al. 2000, Jilma-Stohlawetz et al. 2003, Debili et al. 1990). THBS1 homotrimers contribute to stabilization of the platelet aggregate (Bonnefoy and Hoylaerts 2008). MIR27A is a negative regulator of RUNX1 mRNA translation and may be involved in erythroid/megakaryocytic lineage determination (Ben-Ami et al. 2009).

The RUNX1:CBFB complex stimulates transcription of the PF4 gene, encoding a component of platelet alpha granules (Aneja et al. 2011), the NR4A3 gene, associated with the familial platelet disorder (FPD) (Bluteau et al. 2011), the PRKCQ gene, associated with inherited thrombocytopenia (Jalagadugula et al. 2011), the MYL9 gene, involved in thrombopoiesis (Jalagadugula et al. 2010), and the NFE2 gene, a regulator of erythroid and megakaryocytic maturation and differentiation (Wang et al. 2010).

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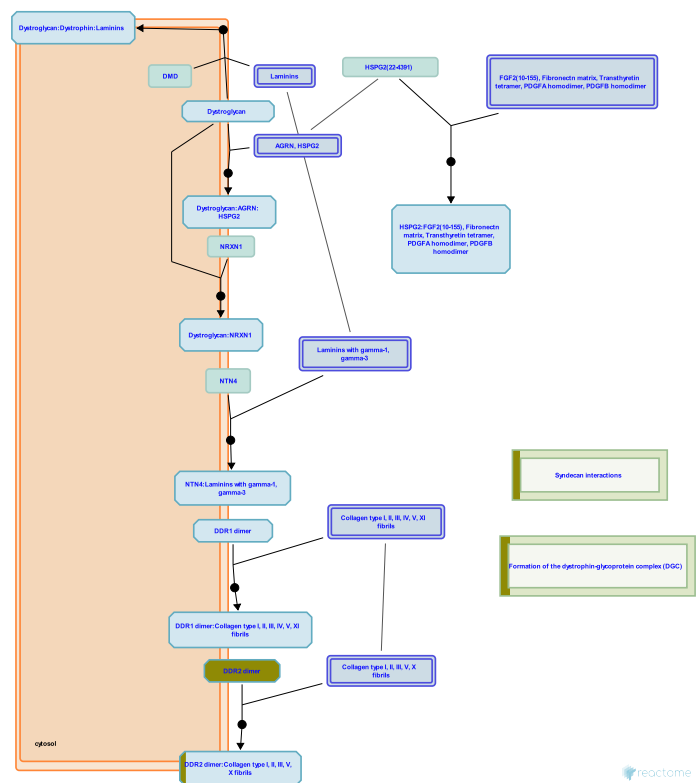
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2016-12-20	Reviewed	Ito Y, Chuang LS
2017-05-09	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
THBS1	P07996	TNRC6B	Q9UPQ9

Input	Ensembl Id
THBS1	ENSG00000137801

11. Non-integrin membrane-ECM interactions ([R-HSA-3000171](#))



Several non-integrin membrane proteins interact with extracellular matrix proteins. Transmembrane proteoglycans may associate with integrins and growth factor receptors to influence their function, or they can signal independently, often influencing the actin cytoskeleton.

References

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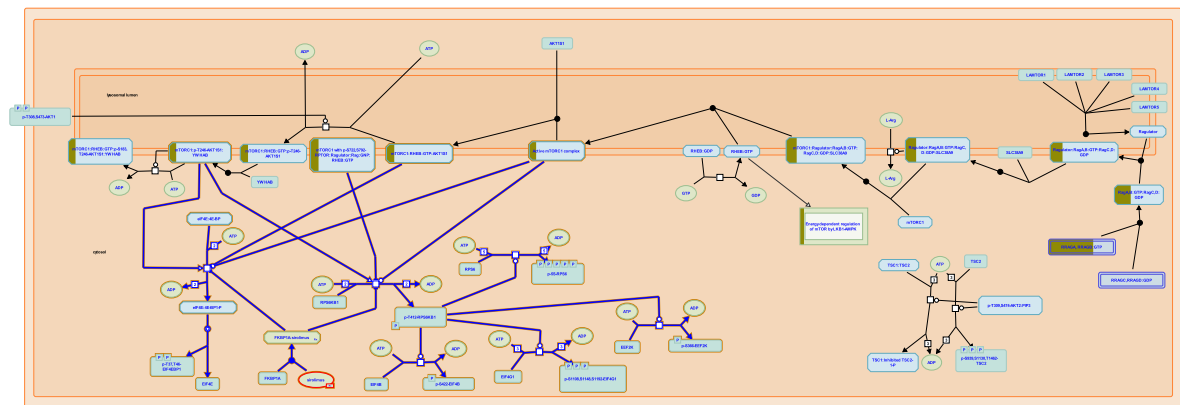
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2013-04-26	Edited	Jupe S
2013-05-22	Reviewed	Ricard-Blum S
2025-11-15	Modified	Weiser JD

3 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
DDR2	Q16832	THBS1	P07996	UTRN	P46939

12. mTORC1-mediated signalling (R-HSA-166208)



reactome

mTORC1 integrates four major signals – growth factors, energy status, oxygen and amino acids – to regulate many processes that are involved in the promotion of cell growth. Growth factors stimulate mTORC1 through the activation of the canonical insulin and Ras signaling pathways. The energy status of the cell is signaled to mTORC1 through AMP-activated protein kinase (AMPK), a key sensor of intracellular energy status (Hardie 2007). Energy depletion (low ATP:ADP ratio) activates AMPK which phosphorylates TSC2, increasing its GAP activity towards Rheb which reduces mTORC1 activation (Inoki et al. 2003). AMPK can reduce mTORC1 activity by directly phosphorylating Raptor (Gwinn et al. 2008). Amino acids positively regulate mTORC1 (reviewed by Guertin & Sabatini 2007). In the presence of amino acids, Rag proteins bind Raptor to promote the relocalization of mTORC1 from the cytoplasm to lysosomal membranes (Puertollano 2014) where it is activated by Rheb (Saucedo et al. 2003, Stocker et al. 2003). Translocation of mTOR to the lysosome requires active Rag GTPases and a complex known as Ragulator, a pentameric protein complex that anchors the Rag GTPases to lysosomes (Sancak et al. 2008, 2010, Bar-Peled et al. 2012). Rag proteins function as heterodimers, consisting of GTP-bound RagA or RagB complexed with GDP-bound RagC or RagD. Amino acids may trigger the GTP loading of RagA/B, thereby promoting binding to raptor and assembly of an activated mTORC1 complex, though a recent study suggested that the activation of mTORC1 is not dependent on Rag GTP charging (Oshiro et al. 2014).

The activity of Rheb is regulated by a complex consisting of tuberous sclerosis complex 1 (TSC1), TSC2, and TBC1 domain family member 7 (TBC1D7) (Huang et al. 2008, Dibble et al. 2012). This complex localizes to lysosomes and functions as a GTPase-activating protein (GAP) that inhibits the activity of Rheb (Menon et al. 2014, Demetriades et al. 2014). In the presence of growth factors or insulin, TSC releases its inhibitory activity on Rheb, thus allowing the activation of mTORC1.

References

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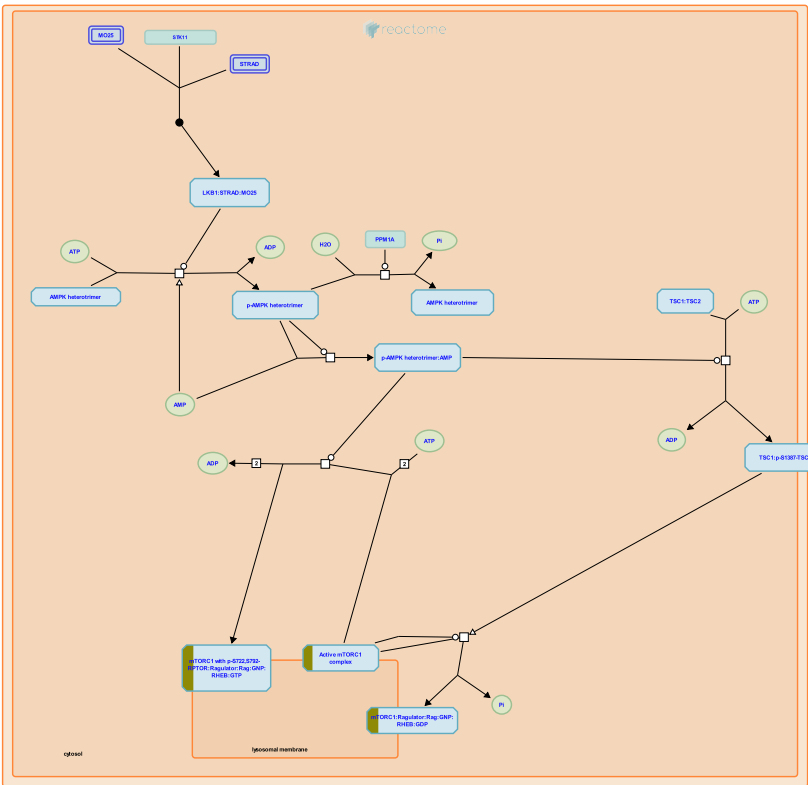
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2015-04-08	Revised	Jupe S
2015-05-14	Reviewed	Zwartkruis FJ

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
RRAGA	Q5VZM2, Q7L523

13. Energy dependent regulation of mTOR by LKB1-AMPK (R-HSA-380972)



Upon formation of a trimeric LKB1:STRAD:MO25 complex, LKB1 phosphorylates and activates AMPK. This phosphorylation is immediately removed in basal conditions by PP2C, but if the cellular AMP:ATP ratio rises, this activation is maintained, as AMP binding by AMPK inhibits the dephosphorylation. AMPK then activates the TSC complex by phosphorylating TSC2. Active TSC activates the intrinsic GTPase activity of Rheb, resulting in GDP-loaded Rheb and inhibition of mTOR pathway.

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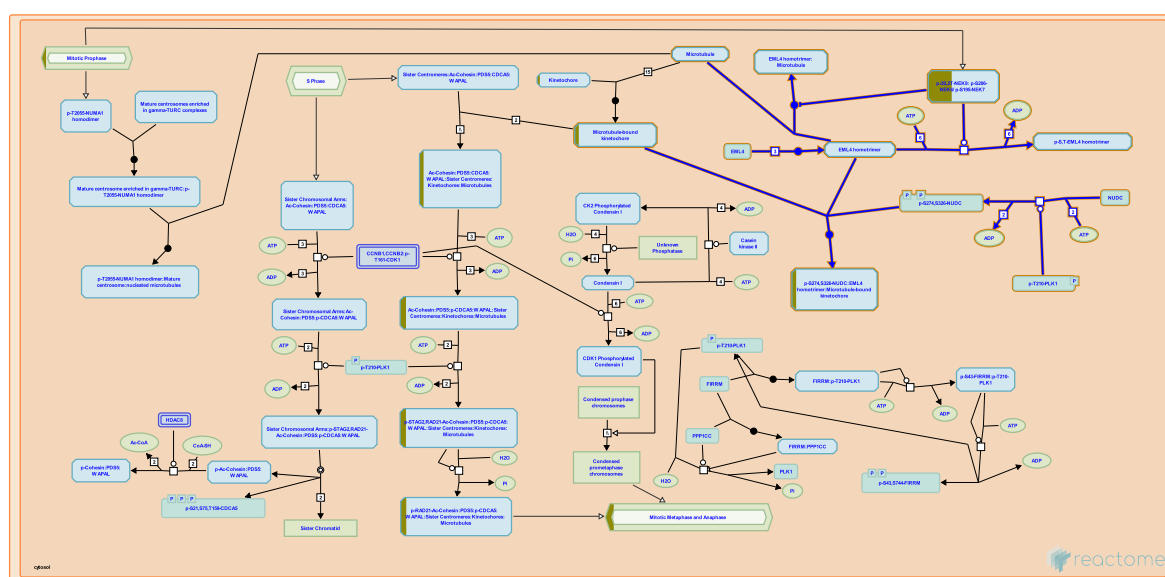
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2008-11-19	Created	Jassal B
2015-04-08	Revised	Jupe S
2015-05-14	Reviewed	Zwartkruis FJ
2025-11-15	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
RRAGA	Q5VZM2, Q7L523

14. EML4 and NUDC in mitotic spindle formation (R-HSA-9648025)



EML4 and NUDC proteins are required for mitotic spindle formation, attachment of spindle microtubule ends to kinetochores, and alignment of mitotic chromosome at the metaphase plate. EML4 is a WD40 family protein that binds to interphase microtubules and stabilizes them (Houtman et al. 2007, Adib et al. 2019). At mitotic entry, EML4 undergoes phosphorylation (Pollmann et al. 2006, Adib et al. 2019) by serine/threonine kinases NEK6 and NEK7, leading to its dissociation from microtubules, which is necessary for the assembly of a dynamic mitotic spindle (Adib et al. 2019). EML4, through its WD40 repeats, interacts with NUDC and recruits it to the kinetochores of the mitotic spindle (Chen et al. 2015). It is possible that other mitotic kinases, besides NEK6 and NEK7, also phosphorylate EML4. Phosphorylation of different residues of EML4 could reduce or increase affinity of EML4 for specific subpopulations of microtubules in mitosis.

A recurrent genomic rearrangement, reported in about 5% cases of non-small cell lung cancer (NSCLC) fuses the N-terminal portion of EML4 with the C-terminal portion of ALK (anaplastic lymphoma kinase), resulting in a constitutively active ALK (Soda et al. 2007, Richards et al. 2015).

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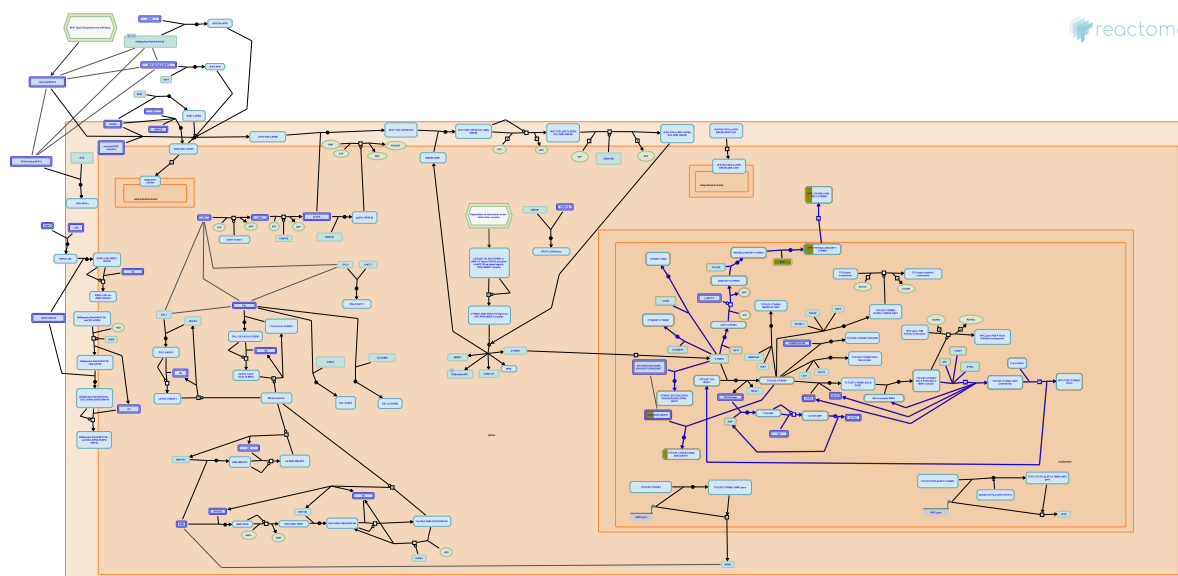
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Date	Action	Author
2019-06-04	Created	Orlic-Milacic M
2019-06-25	Authored	Orlic-Milacic M
2019-09-30	Reviewed	O'Regan L, Fry AM, Lucken KJ
2019-10-03	Reviewed	Bechstedt S
2019-10-07	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

3 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ITGB3BP	Q13352	NEK7	Q8TDX7	XPO1	O14980

15. Deactivation of the beta-catenin transactivating complex (R-HSA-3769402)



The mechanisms involved in downregulation of TCF-dependent transcription are not yet very well understood. beta-catenin is known to recruit a number of transcriptional repressors, including Reptin, SMRT and NCoR, to the TCF/LEF complex, allowing the transition from activation to repression (Bauer et al, 2000; Weiske et al, 2007; Song and Gelmann, 2008). CTNNBIP1 (also known as ICAT) and Chibby are inhibitors of TCF-dependent signaling that function by binding directly to beta-catenin and preventing interactions with critical components of the transactivation machinery (Takemaru et al, 2003; Li et al, 2008; Tago et al, 2000; Graham et al, 2002; Daniels and Weiss, 2002). Chibby additionally promotes the nuclear export of beta-catenin in conjunction with 14-3-3/YWHAZ proteins (Takemura et al, 2003; Li et al, 2008). A couple of recent studies have also suggested a role for nuclear APC in the disassembly of the beta-catenin activation complex (Hamada and Bienz, 2004; Sierra et al, 2006). It is worth noting that while some of the players involved in the disassembly of the beta-catenin transactivating complex are beginning to be worked out in vitro, the significance of their role in vivo is not yet fully understood, and some can be knocked out with little effect on endogenous WNT signaling (see for instance Voronina et al, 2009).

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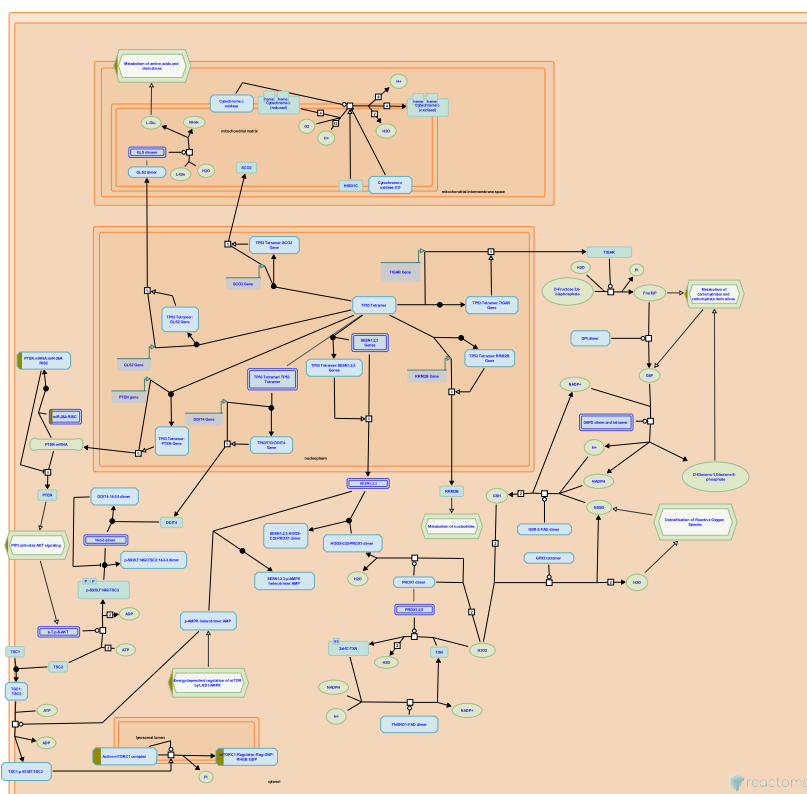
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2013-06-20	Created	Rothfels K
2013-10-03	Edited	Gillespie ME
2014-01-22	Reviewed	Rajakulendran N
2014-02-15	Reviewed	van Amerongen R
2014-04-22	Reviewed	Kikuchi A

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
SOX13	Q9UN79	XPO1	O14980

16. TP53 Regulates Metabolic Genes (R-HSA-5628897)



While the p53 tumor suppressor protein (TP53) is known to inhibit cell growth by inducing apoptosis, senescence and cell cycle arrest, recent studies have found that p53 is also able to influence cell metabolism to prevent tumor development. TP53 regulates transcription of many genes involved in the metabolism of carbohydrates, nucleotides and amino acids, protein synthesis and aerobic respiration.

TP53 stimulates transcription of TIGAR, a D-fructose 2,6-bisphosphatase. TIGAR activity decreases glycolytic rate and lowers ROS (reactive oxygen species) levels in cells (Bensaad et al. 2006). TP53 may also negatively regulate the rate of glycolysis by inhibiting the expression of glucose transporters GLUT1, GLUT3 and GLUT4 (Kondoh et al. 2005, Schwartzberg-Bar-Yoseph et al. 2004, Kawachi et al. 2008).

TP53 negatively regulates several key points in PI3K/AKT signaling and downstream mTOR signaling, decreasing the rate of protein synthesis and, hence, cellular growth. TP53 directly stimulates transcription of the tumor suppressor PTEN, which acts to inhibit PI3K-mediated activation of AKT (Stambolic et al. 2001). TP53 stimulates transcription of sestrin genes, SESN1, SESN2, and SESN3 (Velasco-Miguel et al. 1999, Budanov et al. 2002, Brynczka et al. 2007). One of sestrin functions may be to reduce and reactivate overoxidized peroxiredoxin PRDX1, thereby reducing ROS levels (Budanov et al. 2004, Papadia et al. 2008, Essler et al. 2009). Another function of sestrins is to bind the activated AMPK complex and protect it from AKT-mediated inactivation. By enhancing AMPK activity, sestrins negatively regulate mTOR signaling (Budanov and Karin 2008, Cam et al. 2014). The expression of DDIT4 (REDD1), another negative regulator of mTOR signaling, is directly stimulated by TP53 and TP63. DDIT4 prevents AKT-mediated inactivation of TSC1:TSC2 complex, thus inhibiting mTOR cascade (Cam et al. 2014, Ellisen et al. 2002, DeYoung et al. 2008). TP53 may also be involved, directly or indirectly, in regulation of expression of other participants of PI3K/AKT/mTOR signaling, such as PIK3CA (Singh et al. 2002), TSC2 and AMPKB (Feng et al. 2007).

TP53 regulates mitochondrial metabolism through several routes. TP53 stimulates transcription of SCO2 gene, which encodes a mitochondrial cytochrome c oxidase assembly protein (Matoba et al. 2006). TP53 stimulates transcription of RRM2B gene, which encodes a subunit of the ribonucleotide reductase complex, responsible for the conversion of ribonucleotides to deoxyribonucleotides and essential for the maintenance of mitochondrial DNA content in the cell (Tanaka et al. 2000, Bourdon et al. 2007, Kulawiec et al. 2009). TP53 also transactivates mitochondrial transcription factor A (TFAM), a nuclear-encoded gene important for mitochondrial DNA (mtDNA) transcription and maintenance (Park et al. 2009). Finally, TP53 stimulates transcription of the mitochondrial glutaminase GLS2, leading to increased mitochondrial respiration rate and reduced ROS levels (Hu et al. 2010).

The great majority of tumor cells generate energy through aerobic glycolysis, rather than the much more efficient aerobic mitochondrial respiration, and this metabolic change is known as the Warburg effect (Warburg 1956). Since the majority of tumor cells have impaired TP53 function, and TP53 regulates a number of genes involved in glycolysis and mitochondrial respiration, it is likely that TP53 inactivation plays an important role in the metabolic derangement of cancer cells such as the Warburg effect and the concomitant increased tumorigenicity (reviewed by Feng and Levine 2010). On the other hand, some mutations of TP53 in Li-Fraumeni syndrome may result in the retention of its wild-type metabolic activities while losing cell cycle and apoptosis functions (Wang et al. 2013). Consistent with such human data, some mutations of p53, unlike p53 null state, retain the ability to regulate energy metabolism while being inactive in regulating its classic gene targets involved in cell cycle, apoptosis and senescence. Retention of metabolic and antioxidant functions of p53 protects p53 mutant mice from early onset tumorigenesis (Li et al. 2012).

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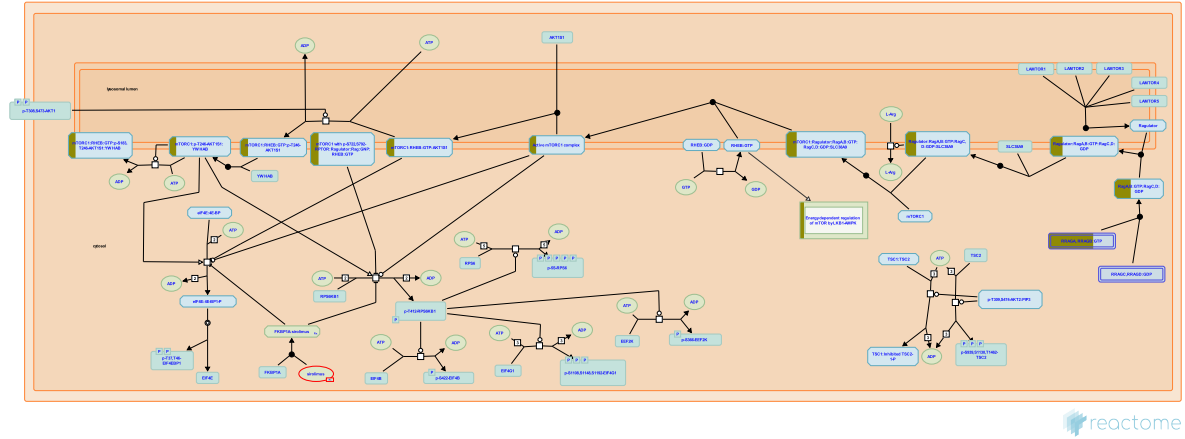
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2014-10-24	Created	Orlic-Milacic M
2014-12-23	Edited	Orlic-Milacic M
2014-12-23	Authored	Orlic-Milacic M
2014-12-30	Reviewed	Hwang PM, Kang JG, Wang PY
2016-02-04	Reviewed	Zaccara S, Inga A
2025-11-15	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 3 Reactome entities

Input	UniProt Id	Input	UniProt Id
RRAGA	Q5VZM2, Q7L523	TNRC6B	Q9UPQ9

17. MTOR signalling (R-HSA-165159)



Target of rapamycin (mTOR) is a highly-conserved serine/threonine kinase that regulates cell growth and division in response to energy levels, growth signals, and nutrients (Zoncu et al. 2011). Control of mTOR activity is critical for the cell since its dysregulation leads to cancer, metabolic disease, and diabetes (Laplante & Sabatini 2012). In cells, mTOR exists as two structurally distinct complexes termed mTOR complex 1 (mTORC1) and mTOR complex 2 (mTORC2), each one with specificity for different sets of effectors. mTORC1 couples energy and nutrient abundance to cell growth and proliferation by balancing anabolic (protein synthesis and nutrient storage) and catabolic (autophagy and utilization of energy stores) processes.

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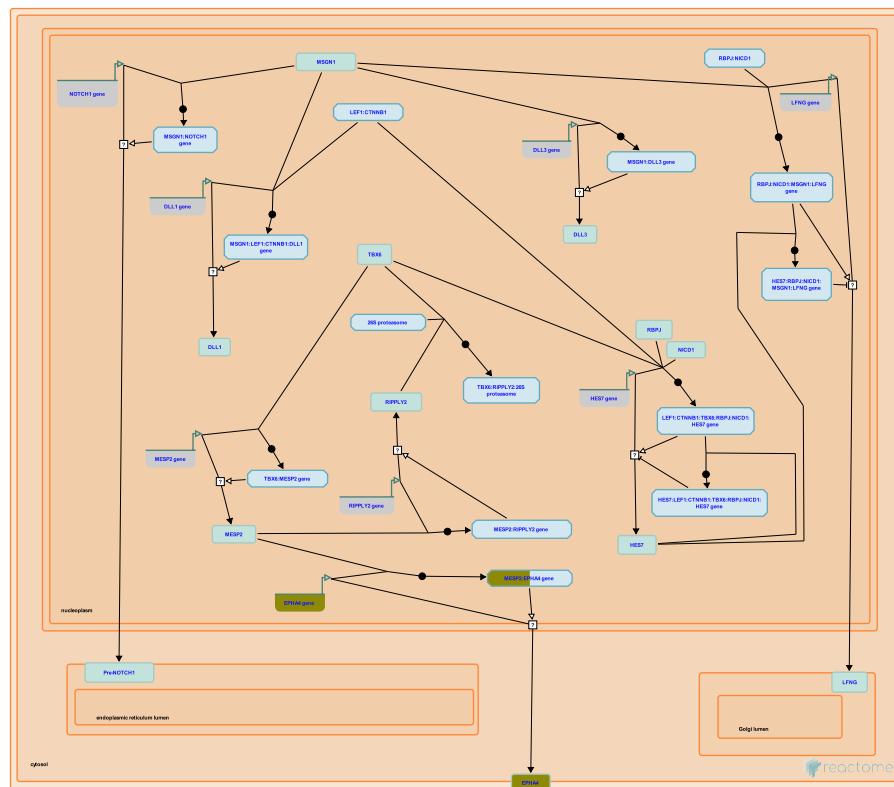
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2005-08-01	Created	de Bono B
2015-01-23	Edited	Jupe S
2015-01-23	Authored	Jupe S
2015-04-08	Revised	Jupe S
2015-05-14	Reviewed	Zwartkruis FJ
2025-11-15	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
RRAGA	Q5VZM2, Q7L523

18. Somitogenesis (R-HSA-9824272)



Somites are bounded segments of mesenchyme that are periodically cleaved, or segmented, from the developing anterior paraxial mesoderm (Diaz-Cuadros et al. 2020). Somite formation is conceptualized using a clock and wavefront model (reviewed in Saga 2012). The clock is present in cells of the presomitic mesoderm and cycles between a permissive state in which segmentation can occur and a refractory state in which it cannot. The wavefront moves along the presomitic mesoderm and causes segmentation where and when it encounters cells in the permissive state, thus the size of the somites is determined by the periodicity of the clock and the migration speed of the wavefront.

The segmentation process is driven by WNT signaling, FGF signaling, and especially the Notch signaling (reviewed in Dunwoodie et al. 2009, Ferjentsik et al. 2009), Hubaud and Pourquie 2014). The intersection of posterior-anterior gradients of WNT and FGF signaling and an anterior-posterior gradient of retinoic acid signaling regulates the position of somite boundaries as perturbation of any of the gradients affects somite boundaries (reviewed in Gibb et al. 2010, Hubard and Pourquie et al. 2014). Thus WNT, FGF, and retinoic acid appear to form the wavefront, also called the determination front. Segmentation periodicity is controlled by HES7-mediated negative feedback loops in the Notch pathway, which constitute a molecular oscillator or segmentation clock (Bessho et al. 2003). Activation of LFNG expression by Notch and inhibition of Notch signaling by LFNG, possibly via regulation of the DLL3 ligand (Bochter et al. 2022), constitute another negative feedback loop that acts as a molecular oscillator (Dale et al. 2003, Falk et al. 2022). Clock oscillations are initiated in nascent presomitic mesoderm in the primitive streak of the gastrulating embryo (Falk et al. 2022) and posterior-to-anterior waves sweep to anterior paraxial mesoderm to regulate MESP2/RIPPLY2 expression to initiate segmentation. MESP2 activates expression of EPHA4 (Nakajima et al. 2006), an Eph receptor that participates in segment boundary formation. MESP2 also activates expression of RIPPLY2 (Morimoto et al. 2007), an inhibitor of TBX6 (Zhao et al. 2018). TBX6 is an activator of MESP2, therefore MESP2 indirectly inhibits its own expression via RIPPLY2.

Mutations in components of the segmentation clock, for example DLL3, MESP2, LFNG, and HES7, cause congenital vertebral defects in humans (Dunwoodie et al. 2009, Nóbrega et al. 2021).

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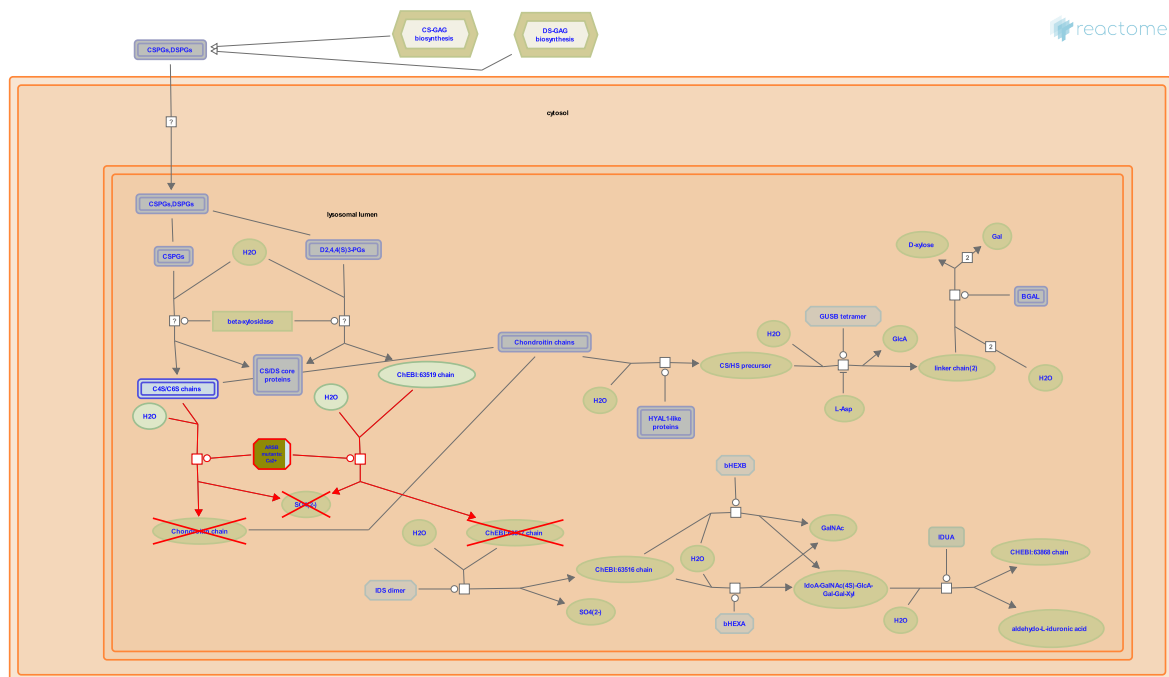
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2023-01-14	Edited	May B
2023-01-14	Authored	May B
2023-01-18	Created	May B
2023-05-09	Reviewed	Yamaguchi TP, Chalamalasetty RB
2023-05-19	Modified	Wright A

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
EPHA4	P54764

Input	Ensembl Id
EPHA4	ENSG00000116106

19. MPS VI - Maroteaux-Lamy syndrome (R-HSA-2206285)



Diseases: mucopolysaccharidosis VI.

Mucopolysaccharidosis type VI (MPS VI, Maroteaux-Lamy syndrome, polydystrophic dwarfism; MIM:253200) is an autosomal recessive lysosomal storage disorder caused by a deficiency in aryl-sulfatase B (ARSB, N-acetyl-galactosamine 4-sulfatase; MIM:611542). It is named after two French physicians, Pierre Maroteaux and Maurice Emil Joseph Lamy. Maroteaux first described this disorder as a novel dysostosis associated with increased urinary excretion of chondroitin sulfate (CS; Maroteaux et al. 1963). The gene encoding ARSB is mapped to chromosome 5q11-q13 (Fidzianska et al. 1984) and contains 8 exons spanning about 206 kb (Karangeorgos et al. 2007). Defective ARSB results in build up of dermatan sulfate (DS) and chondroitin sulfate (CS) in soft tissues causing compression and blockages in blood vessels, nerves, trachea, corneal clouding and disrupting normal bone development. Symptoms are similar to MPS I but with normal intelligence generally (Rapini et al. 2007, Valayannopoulos et al. 2010).

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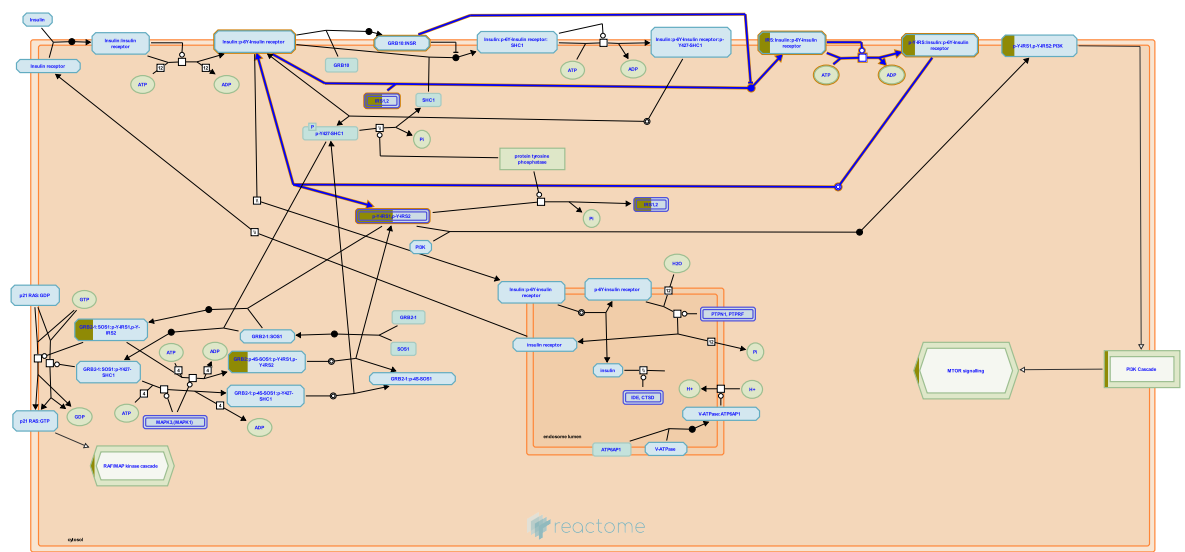
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2012-04-26	Edited	Jassal B
2012-04-26	Authored	Jassal B
2012-04-26	Created	Jassal B
2012-08-27	Reviewed	Alves S, Coutinho MF
2012-08-28	Reviewed	Ashworth J
2025-05-28	Modified	Stephan R

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
ARSB	P15848

20. IRS activation (R-HSA-74713)



IRS is one of the mediators of insulin signalling events. It is activated by phosphorylation and triggers a cascade of events involving PI3K, SOS, RAF and the MAP kinases. The proteins mentioned under IRS are examples of IRS family members acting as indicated. More family members are to be confirmed and added in the future.

Using receptor mutagenesis studies it is known that IRS1 via its PTB domain binds to the insulin receptor at the juxtamembrane region at tyrosine 972. The interaction is further stabilized by the PH domain of IRS1 which interacts with the phospholipids of the plasma membrane. This allows the receptor to phosphorylate IRS1 on up to 13 of its tyrosine residues. Once phosphorylated the IRS1 falls away from the receptor. Now in a tyrosine phosphorylated and hence activated state other proteins can interact with the IRS proteins.

References

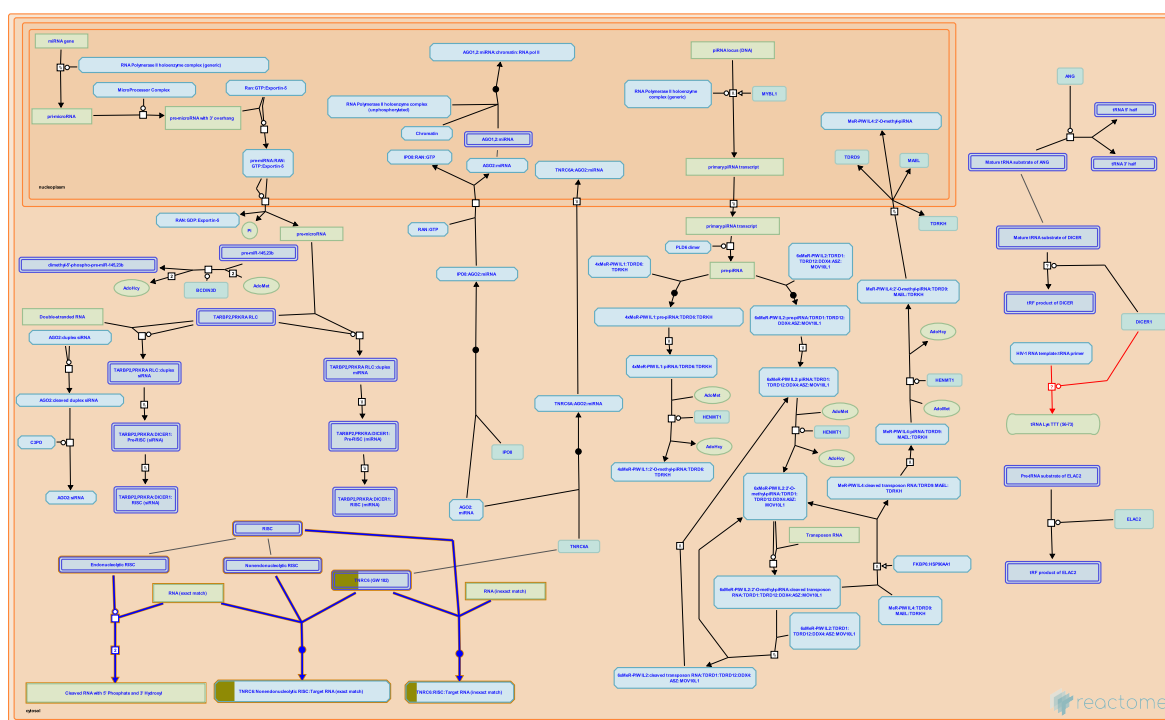
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Date	Action	Author
2003-07-28	Authored	
2003-07-31	Created	Bevan AP

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
IRS2	Q9Y4H2

21. Post-transcriptional silencing by small RNAs (R-HSA-426496)



Cellular compartments: cytosol.

Small RNAs act with components of the RNA-induced silencing complex (RISC) to post-transcriptionally repress expression of mRNAs (reviewed in Nowotny and Yang 2009, Chua et al. 2009). Two mechanisms exist: 1) cleavage of target RNAs by complexes containing Argonaute2 (AGO2, EIF2C2) and a guide RNA that exactly matches the target mRNA and 2) inhibition of translation of target RNAs by complexes containing AGO2 and an inexact matching guide RNA or by complexes containing a nonendonucleolytic Argonaute (AGO1 (EIF2C1), AGO3 (EIF2C3), or AGO4 (EIF2C4)) and a guide RNA of exact or inexact match. Small interfering RNAs (siRNAs) and microRNAs (miRNAs) can serve as guide RNAs in both types of mechanism.

RNAi also appears to direct chromatin modifications that cause transcriptional gene silencing (reviewed in Verdel et al. 2009).

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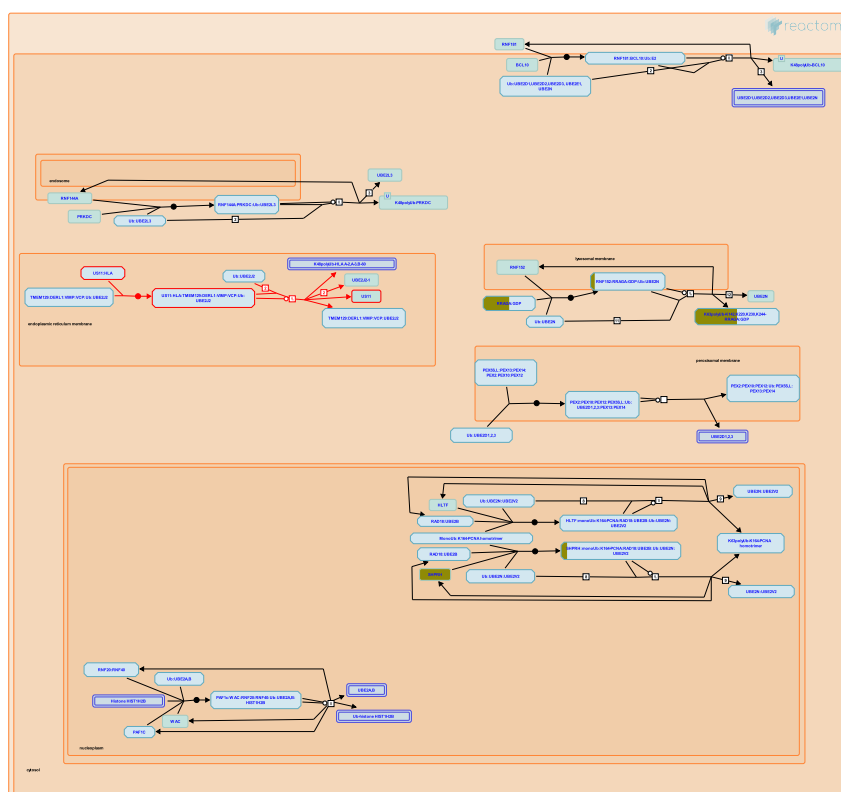
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2009-06-10	Authored	May B
2009-06-17	Created	May B
2012-02-11	Reviewed	Tomari Y
2025-11-15	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
TNRC6B	Q9UPQ9

22. E3 ubiquitin ligases ubiquitinate target proteins (R-HSA-8866654)



E3 ubiquitin ligases catalyze the transfer of an ubiquitin from an E2-ubiquitin conjugate to a target protein. Generally, ubiquitin is transferred via formation of an amide bond to a particular lysine residue of the target protein, but ubiquitylation of cysteine, serine and threonine residues in a few targeted proteins has also been demonstrated (reviewed in McDowell and Philpott 2013, Berndsen and Wolberger 2014). Based on protein homologies, families of E3 ubiquitin ligases have been identified that include RING-type ligases (reviewed in Deshaies et al. 2009, Metzger et al. 2012, Metzger et al. 2014), HECT-type ligases (reviewed in Rotin et al. 2009, Metzger et al. 2012), and RBR-type ligases (reviewed in Dove et al. 2016). A subset of the RING-type ligases participate in CULLIN-RING ligase complexes (CRLs which include SCF complexes, reviewed in Lee and Zhou 2007, Genschik et al. 2013, Skaar et al. 2013, Lee et al. 2014).

Some E3-E2 combinations catalyze mono-ubiquitination of the target protein (reviewed in Nakagawa and Nakayama 2015). Other E3-E2 combinations catalyze conjugation of further ubiquitin monomers to the initial ubiquitin, forming polyubiquitin chains. (It may also be possible for some E3-E2 combinations to preassemble polyubiquitin and transfer it as a unit to the target protein.) Ubiquitin contains several lysine (K) residues and a free alpha amino group to which further ubiquitin can be conjugated. Thus different types of polyubiquitin are possible: K11 linked polyubiquitin is observed in endoplasmic reticulum-associated degradation (ERAD), K29 linked polyubiquitin is observed in lysosomal degradation, K48 linked polyubiquitin directs target proteins to the proteasome for degradation, whereas K63 linked polyubiquitin generally acts as a scaffold to recruit other proteins in several cellular processes, notably DNA repair (reviewed in Komander et al. 2009).

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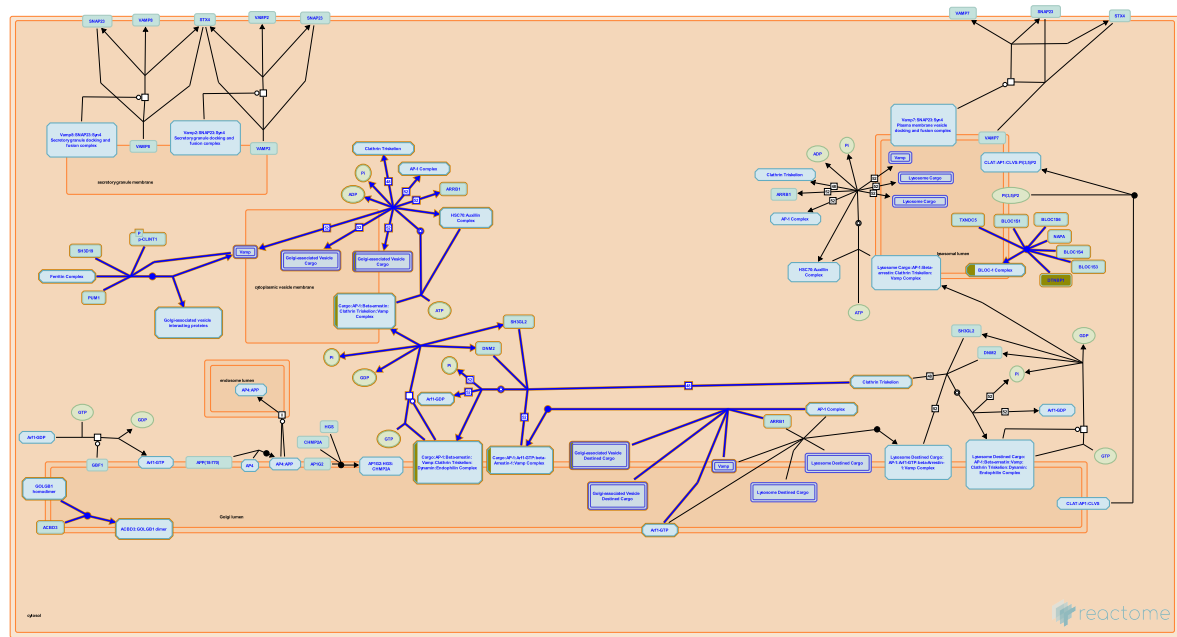
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Date	Action	Author
2016-04-02	Edited	May B
2016-04-02	Authored	May B
2016-04-02	Created	May B
2016-11-03	Reviewed	Azevedo JE

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
RRAGA	Q7L523	SHPRH	Q149N8

23. Golgi Associated Vesicle Biogenesis (R-HSA-432722)



Proteins that have been synthesized, processed and sorted eventually reach the final steps of the secretory pathway. This pathway is responsible not only for proteins that are secreted from the cell but also enzymes and other resident proteins in the lumen of the ER, Golgi, and lysosomes as well as integral proteins transported in the vesicle membranes.

References

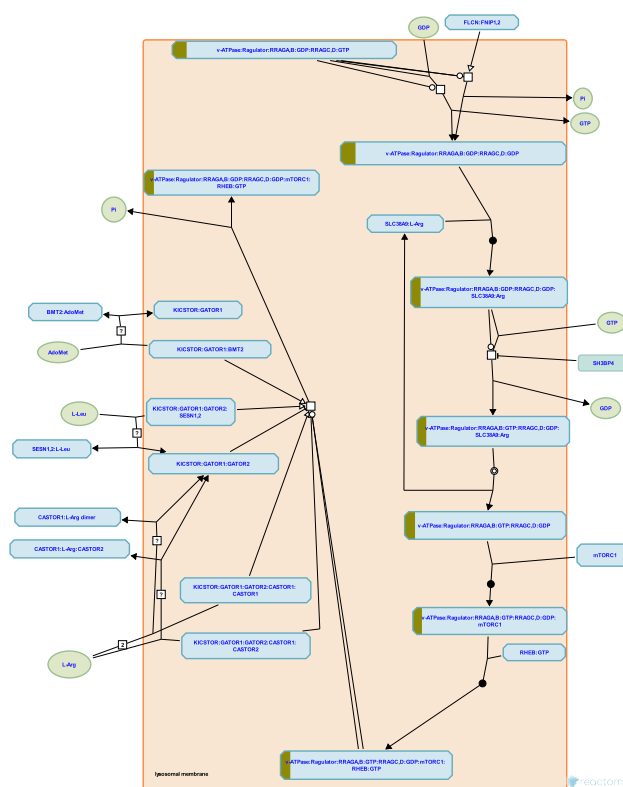
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2009-08-15	Created	Gillespie ME
2009-08-27	Authored	Gillespie ME
2009-08-28	Reviewed	Simpson JC
2025-11-15	Modified	Weiser JD

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
DTNBP1	Q96EV8	SORT1	Q99523

24. Amino acids regulate mTORC1 (R-HSA-9639288)



The mTORC1 complex acts as an integrator that regulates translation, lipid synthesis, autophagy, and cell growth in response to multiple inputs, notably glucose, oxygen, amino acids, and growth factors such as insulin (reviewed in Sabatini 2017, Meng et al. 2018, Kim and Guan 2019).

MTOR, the kinase subunit of mTORC1, is activated by interaction with RHEB:GTP at the cytosolic face of lysosomal membrane (Long et al. 2005, Tee et al. 2005, Long et al. 2007, Yang et al. 2017). Recruitment of mTORC1 to the lysosomal membrane is intricate and incompletely understood. At the center of the system is a complex of two small GTPases, the Rag heterodimer (RRAGA or RRAGB bound to RRAGC or RRAGD). The Rag heterodimer is tethered to the membrane by the Ragulator complex, which also binds the v-ATPase complex. The Rag heterodimer acts as a cross-regulating switch, with the binding of GTP by one subunit inhibiting the exchange of GDP for GTP by the other subunit (Shen et al. 2017). The active conformation of the Rag heterodimer that recruits mTORC1 to the lysosomal membrane is RRAGA,B:GTP:RRAGC,D:GDP while the inactive conformation, RRAGA,B:GDP:RRAGC,D:GTP, releases mTORC1 (Sancak et al. 2008, Kim et al. 2008, Sancak et al. 2010, Lawrence et al. 2018). GTPase activating proteins (GAPs) and guanyl nucleotide exchange factors (GEFs) acting upon the Rag heterodimer thereby regulate recruitment of mTORC1. RHEB:GTP at the lysosomal membrane also binds mTORC1 and directly activates mTORC1. During inactivation of mTORC1 in response to removal of amino acids, the TSC complex, a GAP for RHEB, is required in addition to the inactive Rag complex to release mTORC1 from RHEB and hence fully release mTORC1 from the lysosomal membrane (Demetriades et al. 2014).

Amino acids regulate recruitment of mTORC1 to the lysosomal membrane by at least 4 mechanisms (reviewed in Zhuang et al. 2019, Wolfson and Sabatini 2017, Yao et al. 2017). 1) Sestrin1 (SESN1) or Sestrin2 (SESN2) binds leucine and the Sestrin1,2:leucine complex is then released from the GATOR2 complex, allowing GATOR2 to positively regulate mTORC1 activation (Chantranupong et al. 2014, Parmigiani et al. 2014, Kim et al. 2015, Wolfson et al. 2016, Saxton et al. 2016). 2) CASTOR1 in a homodimer or a heterodimer with CASTOR2 binds arginine and the CASTOR1:arginine complex is likewise released from GATOR2, allowing GATOR2 to activate mTORC1 (Chantranupong et al. 2016, Saxton et al. 2016, Gai et al. 2016, Xia et al. 2016). 3) BMT2 (SAMTOR), a negative regulator of mTORC1 activation, binds S-adenosylmethionine (SAM), a derivative of methionine (Gu et al. 2017). The binding of SAM causes BMT2 to dissociate from GATOR1, allowing the activation of mTORC1. 4) The amino acid transporter SLC38A9 binds arginine and SLC38A9 then acts as a GEF to convert RRAGA,B:GDP to the active form, RRAGA,B:GTP (Rebsamen et al. 2015, Wang et al. 2015, Wyant et al. 2017, Shen and Sabatini 2018). Amino acid starvation also regulates the assembly of the V0 and V1 subunits of v-ATPase by an uncharacterized mechanism (Stransky and Forgac 2015) and v-ATPase is required for activation of mTORC1 by amino acids (Zoncu et al. 2011). Glutamine activates mTORC1 by a mechanism that is independent of the Rag GTPases, requires ARF1, but is not yet fully elucidated (Jewell et al. 2015).

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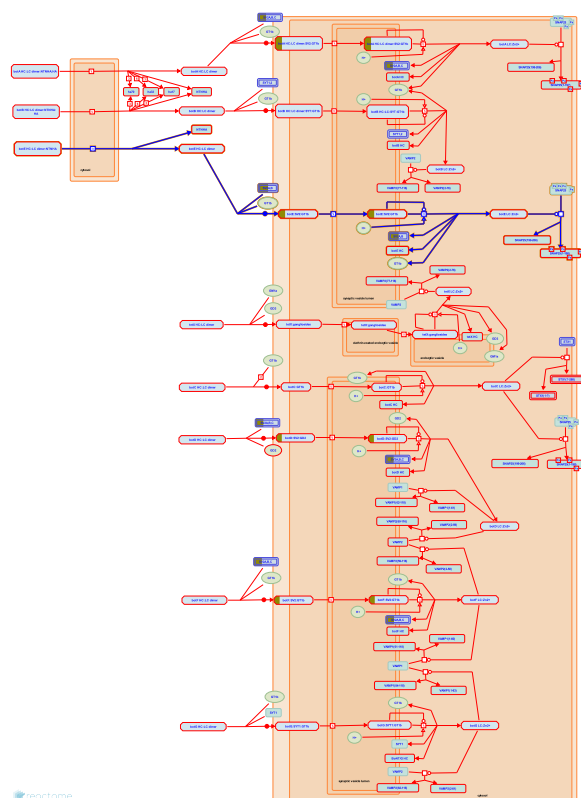
Edit history

Date	Action	Author
2019-03-04	Edited	May B
2019-03-04	Authored	May B
2019-03-09	Created	May B
2019-08-08	Reviewed	Condon KJ, Sabatini DM
2025-11-15	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
RRAGA	Q5VZM2, Q7L523

25. Toxicity of botulinum toxin type E (botE) (R-HSA-5250992)



Diseases: botulism.

Botulinum toxin type E (botE, also known as BoNT/E), a disulfide-bonded heavy chain (HC) - light chain (LC) heterodimer (“dichain”), enters the gut typically as a result of consuming contaminated food (Hatheway 1995), as a complex with nontoxic nonhemagglutinin protein (NTNHA, encoded by the *C. botulinum ntnha* gene) (Benefield et al. 2013). The complex protects the toxin from degradation in the gut and mediates its association with the gut epithelium and transcytosis to enter the circulation (Fujinaga et al. 2013). Circulating toxin molecules associate with gangliosides and synaptic vesicle protein 2 (SV2) exposed by exocytosis at a synapse of a target neuron (Dong et al. 2008; Yowler & Schengrund 2004). Vesicle recycling brings the toxin into the neuron where the vesicle is acidified (Sudhoff 2004). The lowered pH induces a conformational change in the toxin: its HC forms a passage in the vesicle membrane through which its LC is extruded into the neuronal cytosol and released by reduction of the HC - LC disulfide bond (Montal 2010). The LC then catalyzes the cleavage of synaptosome-associated protein 25 (SNAP25) on the cytosolic face of the neuronal plasma membrane (Binz et al. 1994; Schiavo et al. 1993), thereby inhibiting synaptic vesicle fusion with the plasma membrane and exocytosis.

References

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Edit history

Date	Action	Author
2006-06-15	Authored	Krupa S, Gopinathrao G
2007-08-03	Reviewed	Ichtchenko K
2014-02-01	Created	D'Eustachio P
2014-02-11	Revised	D'Eustachio P
2014-02-11	Edited	D'Eustachio P
2014-11-18	Reviewed	Sharma S, Thirunavukkarasu N
2023-10-12	Modified	Weiser JD

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
SV2A	Q7L0J3

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

35 of the submitted entities were found, mapping to 53 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARSB	P15848	CERS6	Q6ZMG9	COPZ2	Q9P299
CRISPLD2	Q9H0B8	CSF1	P09603	DDR2	Q16832
DTNBP1	Q96EV8	EPHA4	P54764	FGD4	Q96M96
FOXP1	Q9H334-8	GBP3	Q9H0R5	GCNT1	Q02742
GEM	P55040	IFNGR1	P15260	IRS2	Q9Y4H2
ITGB3BP	Q13352	METAP2	P50579	NEK7	Q8TDX7
NFIX	Q14938	NIPAL2	Q9H841	NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9
PFKFB3	Q16875	RPLP2	P05387	RRAGA	Q5VZM2, Q7L523
SHPRH	Q149N8	SORT1	Q99523	SOX13	Q9UN79
SV2A	Q7L0J3	THBS1	P07996	TNRC6B	Q9UPQ9
TRNT1	Q96Q11	UTRN	P46939	XPO1	O14980
ZEB1	P37275	ZNF268	Q14587, Q96BR6		

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
CSF1	ENSG00000184371	EPHA4	ENSG00000116106	GBP3	ENSG00000117226
GEM	ENSG00000164949	THBS1	ENSG00000137801	ZEB1	ENSG00000148516

7. Identifiers not found

These 24 identifiers were not found neither mapped to any entity in Reactome.

BAZ2B	BEX2	CAMK2N1	EMP1	ENC1	EPG5	FILIP1L	FRMD4B
GSAP	LPP	NEAT1	PKIA	PLEKHA7	PRICKLE2	RAPH1	SFT2D2
SMCO4	SNN	SNX10	SNX30	SYTL3	TBC1D2B	USP53	ZBTB37